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Stable Matching with Contingent Priorities

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Abstract. Using school choice as a motivating example, we introduce a stylized model of a many-to-one matching market where the clearinghouse aims to implement contingent priorities—priorities that depend on the current assignment—to promote the joint assignment of siblings. We provide a series of guidelines and introduce two natural approaches to implement them: (i) absolute, whereby a prioritized student can displace any student without siblings assigned to the school, and (ii) partial, whereby a prioritized student can only displace students that have a less favorable lottery than their priority provider. We study several properties of the corresponding mechanisms and introduce two variants—soft and hybrid—that relax the enforcement of priorities to ensure existence. Furthermore, we provide mathematical programming formulations to find such stable matching or certify that one does not exist. Finally, using data from the Chilean school choice system, we show that our framework can substantially increase the number of siblings assigned together and reduce the number of unassigned students, all without significant losses in terms of student assignment preferences relative to current practice.

Key words: stable matching, school choice, families, contingent priorities, mathematical programming

1. Introduction

The theory of two-sided many-to-one matching markets, introduced by Gale and Shapley (1962), provides a framework for solving many large-scale real-life matching problems. Examples include entry-level labor markets for doctors and teachers, education markets ranging from daycare to college admissions, and other applications such as refugee resettlement. In many of these markets, the clearinghouse may be interested in finding a stable allocation to guarantee that no coalition of agents has incentives to circumvent the match, while individual agents may care about their matching and that of other agents. For instance, in the hospital-resident problem, couples jointly participate and must coordinate to find two positions that complement each other. In refugee resettlement, agencies may prioritize allocating families with similar backgrounds (e.g., from the same region or speaking the same language) to the same cities. In our primary motivating example, school choice, students may prefer to be assigned with their siblings.

A common approach to accommodate these joint preferences is to provide priorities to increase the chances of specific agents being matched together, such as sibling priorities in school choice.[1]

[1] In refugee resettlement, families may get higher priority in localities where they have relatives based on *family reunification*. This type of priority does not exist in the residency matching problems, as couples must participate together to be considered as such, and candidates do not receive priority (at least explicitly) if their partner already works at a given hospital.

Indeed, between 50% and 60% of countries with a coordinated school choice and assignment system include "family logistics" (siblings) as one of their top priority rules (Neilson 2024).[2] However, most clearinghouses assume that priorities are fixed and known before the assignment process and, thus, cannot accommodate settings in which priorities depend on the current matching. For instance, Boston Public Schools (BPS) prioritizes students who have a sibling currently enrolled for the next academic year—most clearinghouses know this by the time they perform the allocation—but provides no special treatment for families involving multiple applicants participating in the system (e.g., twins, triplets, or siblings applying to different levels). As a result, many families end up being separated, resulting in higher transportation costs, emotional distress, and logistical challenges for parents, among other undesirable outcomes. To tackle this issue, some school districts have introduced special rules for multiples, whereby they try to accommodate them in the same school provided some requirements (e.g., both siblings must submit the same preference list, they must apply to the same level/program, among others).[3] Other clearinghouses, such as the Chilean school choice system, provide sibling priority to applicants if they have a sibling (i) enrolled in the school for the next year or (ii) concurrently participating in the admissions process and assigned in a higher level; nevertheless, they do not consider special treatment of multiples nor flexibility in the direction of priorities. Hence, none of the practical approaches mentioned above fully resolves the problem. Moreover, from a theoretical standpoint, most definitions of stability and justified envy assume fixed and known priorities, offering no clear guidelines on how to handle priorities that depend on the matching. Consequently, the existing theory of stable matching neither fully accounts for these settings nor provides direct solutions.

In this paper, our primary goals are (i) to provide a conceptual framework to incorporate priorities that depend on the current matching, which we refer to as *contingent* priorities; (ii) to design methodologies to find rank-optimal allocations (i.e., that optimize the sum of ranks of the preferences of assignment) that incorporate these priorities; and (iii) to illustrate their benefits and encourage

[2] Many school choice systems justify sibling priority on practical and welfare grounds: it reduces significant logistical burdens for families with multiple children, promotes child well-being by easing transitions, strengthens family–school engagement, and minimizes administrative disruptions from transfer requests (Neilson 2024, Calsamiglia and Güell 2018, UK Department for Education 2021). While this may disadvantage students without siblings in oversubscribed schools, the number of seats affected is typically small relative to the perceived benefits for families, as we illustrate in Section 5. A thorough analysis of alternative policies to enhance fairness toward students without siblings is beyond the scope of this paper.

[3] New York City (NYC) considers a special treatment of multiples in 6th grade (entry level of middle school) starting from the 2022-2023 school year, and it also considers multiples for 3-K and Pre-K (see link for more details). New Orleans (NOLA) uses a unique placement process for multiples, i.e., it is not part of their matching mechanism, and they solve it "manually". Wake County Public Schools (WCPS) goes one step further and only deems feasible matchings where multiples are assigned to the same school.

their adoption in the Chilean school choice system. To this end, we introduce a stylized many-to-one matching model that mirrors the key institutional features of the Chilean school choice system. In our framework, students belong to potentially different grade levels, may have siblings concurrently applying to schools, and each reports strict preferences over available institutions. As mandated by Chilean law, the clearinghouse ranks applicants at each school based on priority groups and uses lotteries to break ties within groups. The mechanism's ultimate objective—also legally required in Chile—is to produce a stable matching that respects priorities while giving precedence to siblings of currently enrolled or newly assigned students. Importantly, these two principles—stability and sibling priority—are not unique to Chile but are central to many school choice systems worldwide and are widely regarded as essential for ensuring fairness, credibility, and family cohesion. Consequently, the framework we propose may be directly applicable to a broad range of school choice systems beyond the Chilean case.

1.1. Contributions

Our work makes several contributions that we now describe in detail.

Framework. Our primary contribution is the introduction and formalization of contingent priorities. To do that, we provide guidelines that delimit their implementation to prevent undesirable outcomes. Namely, we assume that the clearinghouse breaks ties within each group of students using random tie-breakers, that providers of priority must be able to rightfully claim a seat without themselves being prioritized, and that contingent priorities take one of two forms: (i) *absolute*, whereby a prioritized applicant can displace any other student with no siblings assigned to the school; and (ii) *partial*, whereby a prioritized applicant can only displace another with no siblings if the tie-breaker of their provider of priority is better than that of the displaced student. Furthermore, we analyze several properties of the mechanism to find a stable matching for each variant of contingent priorities. In the absolute case, we show that a stable matching under contingent priorities may not exist, that the decision problem of whether there exists such a matching is NP-complete, and that the mechanism to find one is not strategy-proof, although it is strategy-proof in the large. In the partial case, we show that these same properties hold when combined with individual lotteries; however, when combined with family lotteries,[4] the problem is equivalent to having no contingent

[4] Under family lotteries, each student has two random tie-breakers for each school: one shared with their siblings and one individual. The order of students within the same group at each school is determined lexicographically using the family tie-breaker first, followed by the individual tie-breaker. This tie-breaking rule is equivalent to using a single tie-breaker per student per school obtained from adding sufficiently small perturbations to the family tie-breakers (to preserve the order of families). Conversely, under individual lotteries, each student only has their individual tie-breaker, which is independent of that of their siblings.

priorities, inheriting all the properties of the standard case. Finally, we show that our proposed framework is transparent and verifiable, and we introduce two variants—soft and hybrid—that relax the enforcement of priorities to mitigate some of the limitations mentioned above.

Formulations. We provide mathematical programming formulations that enable us to either find a stable matching for each type of contingent priority (i.e., absolute and partial) or show infeasibility. Moreover, our formulations are flexible enough to accommodate different objectives and several practical concerns, including the implementation of hard, soft or hybrid priorities, sibling-based priorities coming from students already enrolled, and secured enrollment for current students seeking to transfer to another school, among others. Finally, we present novel mathematical programming formulations to (i) find a stable matching under contingent priorities in the absence of lotteries, and (ii) identify a stable matching that maximizes the number of siblings assigned together under the standard notion of stability (i.e., without contingent priorities).

Impact. To demonstrate the benefits of our framework and push for a policy change, we apply it to the Chilean school choice system and compare it to key benchmarks, including the student-optimal stable matching under the standard concept of stability and the mechanism currently used in Chile. First, we empirically show that enforcing absolute priorities as a hard requirement increases the number of siblings assigned together relative to all benchmarks, while preserving or improving other assignment outcomes. However, this approach faces the potential drawback of the non-existence of a stable matching. The soft version guarantees existence but offers limited benefits compared to current practice. Notably, a hybrid approach that combines soft priorities with minimum requirements on the number of effective providers strikes a balance between feasibility and increasing the number of siblings assigned together. Second, we demonstrate that the standard notion of stability is inadequate for increasing the number of siblings placed together, as the differences between the stable matching that maximizes sibling placement and the student-optimal matching are non-existent. Third, we evaluate two additional policy changes aiming to further increase the joint assignment of siblings: (i) modifying the objective function to maximize the number of priority receivers and (ii) removing initial lotteries. We find that the latter can significantly increase the number of siblings assigned together. Finally, we test the robustness of our results across regions and tie-breaking rules, finding consistent patterns across regions but notable variation across tie-breaking schemes. These results suggest that the choice of contingent priorities and tie-breaking rules should be tailored to the decision-maker's policy goals.

1.2. Organization

The remainder of this paper is organized as follows. In Section 2, we discuss the relevant literature. In Section 3, we introduce our model and discuss several properties of the mechanisms to find stable matchings under contingent priorities. In Section 4, we provide mathematical programming formulations to identify them. In Section 5, we illustrate the potential benefits of our framework using data from the Chilean school choice system. Finally, in Section 6, we conclude.

2. Literature

Our paper is related to several strands of the literature.

Matching with families. A recent strand of the literature has extended the classic school choice model (Abdulkadiroğlu and Sönmez 2003) to incorporate families. Dur et al. (2022) consider a setting where siblings report the same preferences, and assignments are feasible if and only if all family members are assigned to the same school (or all of them are unassigned). Correa et al. (2022) also consider a model with siblings applying to potentially different levels. To prioritize the joint assignment of siblings, Correa et al. (2022) introduce (i) the use of lotteries at the family level; (ii) a heuristic that processes levels sequentially in decreasing order, updating priorities in each step to capture siblings' priorities that result from the assignment of higher levels; and (iii) the option for families to report that they prefer their siblings to be assigned to the same school rather than following their individual reported preferences. Sun et al. (2023) define rationality and stability in a daycare matching model where siblings may seek transfers to more preferred daycares and families can express joint preference lists. Finally, Sun et al. (2025) provide conditions for the existence of stable assignments that support the joint placement of siblings in large random markets.

Matching with couples. Our paper also relates to the literature on matching with couples, often motivated by labor markets such as the medical residency match. When couples participate, stable matchings may not exist (Roth 1984), and the decision problem of determining whether one exists is NP-complete (Ronn 1990, McDermid and Manlove 2010, Biró et al. 2011). However, existence can be ensured under certain conditions—for example, if preferences are weakly responsive (Klaus and Klijn 2005) or if capacities are slightly expanded (Nguyen and Vohra 2018)—or can occur with high probability in large markets under specific assumptions (Kojima et al. 2013, Ashlagi et al. 2014). Beyond existence and tractability, the literature has focused on competing notions of stability, with three prevalent ones: MM-stability (McDermid and Manlove 2010), BIS-stability (Biró et al. 2011), and KPR-stability (Kojima et al. 2013). See Biró and Klijn (2013) for a comprehensive survey and Appendix A.1 for a discussion of how these definitions relate to our setting.

Depending on the notion of stability, several approaches have been proposed to find a stable matching when one exists, including extensions of the Deferred Acceptance algorithm (Roth and Peranson 1999, Biró et al. 2011, Ashlagi et al. 2014, Kojima et al. 2013), genetic algorithms (Aldershof and Carducci 1999), local search (Marx and Schlotter 2011), SAT-based methods (Drummond et al. 2015), and other heuristics. Among these, the most relevant to our work are mathematical programming approaches. Biró et al. (2014) presents an integer programming formulation for finding a MM-stable matching of maximum size, while Delorme et al. (2021) introduce a unified model that, along with minor changes, can capture the three notions of stability mentioned above.

Matching with complementarities. Beyond families and couples, the matching literature has studied other settings with complementarities. For instance, Ashlagi and Shi (2014) show that correlating lotteries can increase community cohesion by increasing the probability of neighbors going to the same schools. Dur and Wiseman (2019) also study the matching problem with neighbors and show that a stable matching may not exist if students have preferences over joint assignments with their neighbors. Kamada and Kojima (2015) study matching markets where the clearinghouse cares about the composition of the match and, thus, imposes distributional constraints. Nguyen and Vohra (2019) also study the problem with distributional concerns but consider these constraints as soft bounds and provide ex-post guarantees on how close the constraints are satisfied while preserving stability. Nguyen et al. (2021) introduce a new model of many-to-one matching where agents with multi-unit demand maximize a cardinal linear objective subject to multidimensional knapsack constraints. Finally, motivated by labor markets, Dooley and Dickerson (2020) and Knittel et al. (2022) study the "affiliate matching problem", in which firms have preferences over the applicants for their positions but also over the placement of their own workers.

3. Model

We now introduce a two-sided matching market model that includes a priority system. To facilitate the exposition, we use school choice with sibling priorities as a concrete application of the model.

Students. Let $\mathcal{S}$ be a finite set of students and let $\mathcal{F} \subseteq 2^{\mathcal{S}}$ be a partition of $\mathcal{S}$; we refer to each $f \in \mathcal{F}$ as a *family* and denote its size by $|f|$. Given two students s and s', we say that they are *siblings* if there is a family $f \in \mathcal{F}$ such that $s, s' \in f$. If $f \in \mathcal{F}$ is such that $f = \{s\}$, then we say that s has no siblings. With a slight abuse of notation, we define the function $f : \mathcal{S} \to \mathcal{F}$ that maps a student into their specific family, i.e., each student $s \in \mathcal{S}$ belongs to family $f(s) \in \mathcal{F}$. Consequently, students s and s' are siblings if $f(s) = f(s')$.

Levels. Let $\mathcal{L}$ be a finite set of levels (e.g., from Pre-K to 12th grade). With a slight abuse of notation, we define the function $\ell : \mathcal{S} \to \mathcal{L}$ that maps each student $s \in \mathcal{S}$ into the level $\ell(s)$ they belong to. Moreover, we denote by $\mathcal{S}^\ell \subseteq \mathcal{S}$ the set of students that belong to level $\ell \in \mathcal{L}$; thus, the collection of sets $\left\{\mathcal{S}^\ell\right\}_{\ell \in \mathcal{L}}$ also defines a partition over $\mathcal{S}$.

Schools. Let C be a finite set of schools. We assume that each school $c \in C$ offers $q_c^\ell \in \mathbb{Z}_+$ seats on level $\ell \in \mathcal{L}$, where $q_c^\ell = 0$ means that school c does not offer level ℓ. When clear from the context, we extend the set of schools to include the outside option of being unassigned, which we denote by $\emptyset$ and whose capacity in each level we assume to be sufficiently large (e.g., $q_\emptyset^\ell = \left|\mathcal{S}^\ell\right|$), so every student can potentially be unassigned.

Preferences. Let $\succ_s$ be the strict preference order of student s over schools in $C \cup \{\emptyset\}$. Then, we use (i) $c \succ_s c'$ to represent that student s *strictly prefers* school c over school c', (ii) $c \succeq_s c'$ to capture that s *weakly prefers* c over c', and (iii) $\emptyset \succ_s c$ to denote that s prefers to be unassigned over attending school c. With a slight abuse of notation, we use $|\succ_s|$ to represent the number of schools that student s strictly prefers over being unassigned, i.e., $|\succ_s| = |\{c \in C \, : \, c \succ_c \emptyset\}|$.

Priorities. Let $\mathcal{G}$ be a finite set of groups that schools use to prioritize students, and let $g : \mathcal{S} \times C \to \mathcal{G}$ be the function that maps each pair $(s,c) \in \mathcal{S} \times C$ to a group $g(s,c) \in \mathcal{G}$. For simplicity, we assume $\mathcal{G} = \{1, \ldots, |\mathcal{G}|\}$, where students with lower values of $g(\cdot, c)$ are strictly preferred over those with higher values for all $c \in C$ and, thus, $g(s,c) = |\mathcal{G}|$ corresponds to students with no priority.[5] Additionally, let $\boldsymbol{p} \in \mathbb{R}_+^{\mathcal{S} \times C}$ be the vector of random tie-breakers (lotteries) that the clearinghouse uses to resolve ties among students within the same group, i.e., $p_{s,c} \in \mathbb{R}_+$ represents the random tie-breaker of student s at school c. As we later discuss, we consider two different tie-breaking rules: (i) at the *family level*, where all family members are initially assigned the same tie-breaker, with slight perturbations introduced afterward to break ties within the family while preserving the overall family order, and (ii) at the *individual level*, where each student is independently assigned a different random tie-breaker. Finally, the combination of groups and random tie-breakers defines a unique lexicographic priority order $\succ_c$ for each school $c \in C$, which we refer to as *initial priority order*. Specifically, for any $s, s' \in \mathcal{S}$, we say that $s \succ_c s'$ if (i) $g(s,c) < g(s',c)$, or (ii) $g(s,c) = g(s',c)$ and $p_{s,c} < p_{s',c}$ for any $s, s' \in \mathcal{S}$; that is, groups take precedence over lotteries. In the remainder of the paper, we say that a student s has higher priority than s' at school c if $s \succ_c s'$ and, similarly, we say that groups with lower values in $\mathcal{G}$ have higher priority.

[5] Priority groups are defined in advance by the clearinghouse based on student attributes such as residence in a school's walk-zone, sibling enrollment, or socioeconomic disadvantage. This group mapping serves purely as a labeling system, where lower-numbered groups represent higher priority. A special case occurs when all students belong to the same group, as in the no-priority setting.

Matching. Let $\mathcal{V} \subseteq \mathcal{S} \times (C \cup \{\emptyset\})$ be the set of feasible pairs, i.e., $(s,c) \in \mathcal{V}$ implies that $c \succeq_s \emptyset$ and $q_c^{\ell(s)} > 0$. A matching is an assignment $\mu \subseteq \mathcal{V}$ such that (i) each student is assigned to at most one school in C, and (ii) each school is assigned at most its capacity in each level. Formally, for $\mu \subseteq \mathcal{V}$, let $\mu(s) \in C \cup \{\emptyset\}$ be the school that student s was assigned to, $\mu(c) \subseteq \mathcal{S}$ be the set of students assigned to school c, and $\mu^\ell(c)$ be the set of students assigned to school c in level $\ell \in \mathcal{L}$. Then, a matching satisfies that (i) $\mu(s) \in C \cup \{\emptyset\}$ for all students $s \in \mathcal{S}$ and (ii) $\left|\mu^\ell(c)\right| \leq q_c^\ell$ for all schools $c \in C$ and levels $\ell \in \mathcal{L}$.[6]

Stability. As Roth (2002) discusses, a desirable property of any matching is stability, i.e., that there is no group of agents that prefer to circumvent their current match and be matched to each other. To distinguish it from our notion of contingent stability that we define later, we call it *initial* stability. Formally, given a matching $\mu \subseteq \mathcal{V}$, we say that student s has *initial justified envy* towards another student s' assigned to school c if (i) $\ell(s) = \ell(s')$, (ii) $c \succ_s \mu(s)$, and (iii) $s \succ_c s'$. In words, the first condition states that both students belong to the same level; the second condition implies that student s prefers school c rather than $\mu(s)$; and the third condition states that student s has higher priority than s' in school c. In addition, we say that a matching μ is *non-wasteful* if there is no student $s \in \mathcal{S}$ and school $c \in C$ such that $c \succ_s \mu(s)$ and $\left|\mu^\ell(c)\right| < q_c^\ell$. Finally, we say that a matching is *initially stable* if it is *non-wasteful* and no student has *initial justified envy*.

3.1. Contingent Priorities

Many school districts heuristically prioritize students based on the matching of their siblings who are concurrently seeking admission to a new school. To accomplish this, most school districts either (i) process siblings applying to the same level manually (outside the primary matching mechanism), as in NYC and NOLA, or (ii) define an arbitrary order among levels and process them sequentially using the deferred acceptance algorithm, updating school priorities (to account for the sibling priorities arising from the levels processed thus far as an additional group) before moving to the next level, as it is the case in Chile (Correa et al. 2022). These solutions face several limitations, including limited scalability, reach, arbitrariness in prioritization (as discussed in Appendix D.1.1), and often have a limited impact, failing to effectively increase the joint assignment of siblings.

This section aims to provide a framework to incorporate *contingent* priorities in a general and scalable way. To this end, we start by formalizing the concept of providing *contingent priority*.

[6] Notice that the model captures other single-level applications such as refugee resettlement, college admissions and the hospital-resident problem.

DEFINITION 1 (CONTINGENT SIBLINGS PRIORITY). Given a matching μ, a student s provides contingent priority in school c if $\mu(s) = c$ and there exists a student $s' \in \mathcal{S} \setminus \{s\}$ such that (i) $f(s) = f(s')$ and (ii) $\mu(s') \preceq_{s'} c$.

Based on this definition, a student s provides contingent priority to their siblings in school c if they are assigned to it in the current matching μ and one of their siblings may benefit from it. Note, however, that Definition 1 does not specify how *receiving* priority affects schools' (initial) priority orders. For example, clearinghouses could implement contingent priorities by either (i) modifying the set of initial groups by creating an additional one that corresponds to students with contingent priority, somewhat similar to the approach implemented in Chile; or (ii) updating the random tie-breakers of the students with siblings assigned to the school, in the same spirit of admissions systems that process multiples as a batch—with all their members having the same lottery and priority—demanding more than one seat. Both approaches differ in how much they prioritize the joint assignment of siblings relative to sticking to the initial priority order $\succ_c$ given by the initial groups and random tie-breakers. The first approach, which we refer to as *absolute* contingent priority, provides full priority to students with siblings, as they can displace any student without priority. The second approach, which we refer to as *partial* contingent priority, provides moderate priority to students with siblings, as they can only displace students with worse random tie-breakers than their priority provider. Both approaches are formally defined in Section 3.2.

The inclusion of contingent priorities introduces several challenges. First, as shown in Example 2 in Appendix A.2, the ordering of students across schools is no longer unique, as it may change depending on their siblings' (current) matching. Consequently, we cannot apply the notion of initial stability (as defined earlier) or use the deferred acceptance algorithm, as both require fixed and known priority orders. Similarly, we cannot use algorithms that permanently reject students in each iteration (as in the mechanism used by the NRMP to solve the matching with couples problem or the Immediate Acceptance algorithm) because contingent priorities may change as the matching process progresses and, thus, initially rejected students may later get prioritized. Finally, multiple matchings may account for contingent priorities, so there may not be a unique one that is Pareto-dominant for students. As a result, identifying the "best" stable matching under contingent priorities may require defining a clear objective and exploring various methods to find the optimal solution. In the following sections, we address each of these challenges.

3.2. Refining Contingent Priorities

There are technical nuances that must be addressed to effectively define the concept of providing contingent priority without running into "undesirable" matchings, as Example 1 illustrates.

EXAMPLE 1. Consider an instance with a set of students $\mathcal{S} = \{s_1, s_2, s_3, f_1, f_2, f_1', f_2'\}$ where $f = \{f_1, f_2\}$ and $f' = \{f_1', f_2'\}$ are siblings, and a single school c with capacity 4 and a single level. Moreover, suppose that every student initially belongs to the same group, i.e., $g(s,c) = g$ for all $(s,c) \in \mathcal{V}$ and that the initial random-tie breakers in school c are such that $p_{s_1,c} < p_{s_2,c} < p_{s_3,c} < p_{f_1,c} < p_{f_1',c} < p_{f_2,c} < p_{f_2',c}$. Then, one possible matching is

$$\mu = \{(s_1,c),(s_2,c),(s_3,c),(f_1,c),(f_2,\emptyset),(f_1',\emptyset),(f_2',\emptyset)\}.$$

However, the alternative matchings

$$\mu' = \{(s_1,\emptyset),(s_2,\emptyset),(s_3,\emptyset),(f_1,c),(f_2,c),(f_1',c),(f_2',c)\}$$
$$\mu'' = \{(s_1,c),(s_2,c),(s_3,\emptyset),(f_1,c),(f_2,c),(f_1',\emptyset),(f_2',\emptyset)\}$$

are also feasible in terms of capacity and differ in how students with siblings are prioritized.
The matching μ' may not be desirable since neither f_1' nor f_2' would be matched to school c without contingent priority, so it may be "unfair" to allow them to skip everyone else in the initial ordering. This differs from μ'' since f_1 would have also been matched to c in the absence of contingent priority and, consequently, could potentially provide contingent priority to f_2. □

To prevent undesirable matchings like the one discussed in Example 1, we refine the definition of who can provide contingent siblings priority in Definition 2.

DEFINITION 2 (CONTINGENT SIBLING PRIORITY, REFINED). Given a matching μ, a student s provides contingent priority in school c if $\mu(s) = c$ and there exists a student $s' \in \mathcal{S} \setminus \{s\}$ such that (i) $f(s) = f(s')$, (ii) $\mu(s') \preceq_{s'} c$, and (iii) $\left|\{s'' \in \mathcal{S}^{\ell(s)} : s'' \succ_c s,\ c \succeq_{s''} \mu(s'')\}\right| \leq q_c^{\ell(s)} - 1$.

The first two conditions coincide with Definition 1, requiring that the students involved are siblings and that the potential receiver may benefit from the contingent priority. The last condition (iii) restricts priority providers to students who could rightfully claim a spot in school c under the initial priority order and matching μ, meaning that the number of students with higher priority who weakly prefer c is below its capacity.[7] This additional requirement (relative to Definition 1) is

[7] As illustrated in Example 3 in Appendix A.2, this condition is not equivalent to requiring that (s,c) belongs to some initially stable matching (i.e., under $\succ_{c'}$ for all $c' \in C$), since we allow siblings to potentially benefit from changes in the priority orders of other schools triggered by contingent priorities.

motivated by fairness and legal considerations raised in discussions with policymakers. Without it, students with no realistic chance of admission (e.g., due to an unfavorable lottery number) could act as providers of priority, enabling families to entirely circumvent the tie-breaking process—legally mandated in settings such as Chile. Condition (iii) restricts the provision of priority to students who could plausibly gain admission on their own, thereby preventing its wholesale override of the initial ordering. That said, in contexts where these concerns are less relevant, alternative formulations of condition (iii) could be considered. For instance, in Section 5.3.2, we study a policy variation that removes initial lotteries altogether, effectively eliminating condition (iii) and allowing any student with siblings to act as a priority provider. Other potential approaches include setting aside reserves for receivers and providers or boosting priorities when two siblings list the same school.

In the remainder of this paper, we assume that the action of *receiving contingent priority* in a given matching μ is captured by the *contingent priority order* $\succ^{\mu}$, which is uniquely defined by (i) a function $g^{\mu} : \mathcal{S} \times C \to \mathcal{G}^{\mu}$ that maps each pair $(s, c) \in \mathcal{S} \times C$ to their *contingent group* in $\mathcal{G}^{\mu}$, where $\mathbb{Z}_{+} \supseteq \mathcal{G}^{\mu} \supseteq \mathcal{G}$, and (ii) a vector $\boldsymbol{p}^{\mu} \in \mathbb{R}_{+}^{\mathcal{S}\times C}$ of *contingent tie-breakers*, both of which depend on their initial counterparts g and $\boldsymbol{p}$. Then, we use $s \succ_{c}^{\mu} s'$ to represent that s has higher contingent priority than s' in school c, which holds if either (i) $g^{\mu}(s,c) < g^{\mu}(s',c)$ or (ii) $g^{\mu}(s,c) = g^{\mu}(s',c)$ and $p_{s,c}^{\mu} < p_{s',c}^{\mu}$. Furthermore, we use $z^{\mu}(s,c) = 1$ to capture that student s has the most favorable tie-breaker among the members of their family who provide contingent priority in school c as in Definition 2—in which case we say that s is the *effective priority provider* for their family in school c—, and $z^{\mu}(s,c) = 0$ otherwise. As a result, note that (i) $\sum_{s\in f} z^{\mu}(s,c) \leq 1$ for all $f \in \mathcal{F}$ and $c \in C$, i.e., each family has at most one effective provider of contingent priority in each school, and (ii) $z^{\mu}(s,c) = 0$ for all s such that $f(s) = \{s\}$ or $(s',c) \notin \mathcal{V}$ for all $s' \in f(s) \setminus \{s\}$, i.e., students with no siblings applying to school c cannot provide contingent priority in that school.

Example 1 (revisited) Note that μ'' and μ satisfy Definition 2 but differ in how a prioritized student compares to another without priority and how they implement contingent priorities. On the one hand, μ'' is the resulting matching if contingent priorities allow prioritized students to displace any student without priority, which can be implemented by considering students with contingent priority as an additional group with higher priority. For instance, the matching μ'' can be obtained considering $g^{\mu''}$ such that $g^{\mu''}(f_1,c) = g^{\mu''}(f_2,c) < g^{\mu''}(s,c)$, $\forall s \in \mathcal{S} \setminus f$, enabling student f_2 to displace s_3 in school c. On the other hand, μ is the resulting matching if contingent priorities allow prioritized students to displace others with worse random tie-breakers than their priority provider, which could be captured by the following $\boldsymbol{p}^\mu$: $p^\mu_{s_1,c} < p^\mu_{s_2,c} < p^\mu_{s_3,c} < p^\mu_{f_1,c} = p^\mu_{f_2,c} < p^\mu_{f_1',c} < p^\mu_{f_2',c}$. With these lotteries, note that f_1's tie-breaker does not help f_2 to displace s_1, s_2 and s_3. □

Both approaches discussed above differ in how much they prioritize the joint assignment of siblings relative to sticking to the initial priority order $\succ_c$ given by the initial groups and random tie-breakers. Under the first approach, which we refer to as *absolute* contingent priority, students with siblings are promoted to a higher priority group, granting them full precedence over all students without sibling priority, independent of lottery realizations. Under the second approach, which we refer to as *partial* contingent priority, sibling-linked applicants retain their initial priority group but inherit the lottery number of their priority provider; as a result, they can only displace students within the same priority group who have weaker random tie-breakers. We formally introduce these two types of *contingent* priority in Definitions 3 and 4, respectively.

Definition 3 (Absolute Contingent Priority). Contingent priorities are *absolute* if, given a matching μ, contingent lotteries and groups are such that, for any $(s,c) \in \mathcal{V}$, $p^\mu_{s,c} = p_{s,c}$ and

$$g^\mu(s,c) = \begin{cases} g(s,c) & \text{if } g(s,c) < |\mathcal{G}| \\ g(s,c) & \text{if } \left(z^\mu(s,c) = 1 \text{ and } |f(s) \cap \mu(c)| \geq 2\right) \\ & \quad \text{or } \left(\exists s' \in f(s) \setminus \{s\} \text{ with } z^\mu(s',c) = 1\right) \\ |\mathcal{G}| + 1 & \text{otherwise.} \end{cases} \tag{1}$$

Definition 4 (Partial Contingent Priority). Contingent priorities are *partial* if, given a matching μ, contingent lotteries and groups are such that, for each $(s,c) \in \mathcal{V}$, $g^\mu(s,c) = g(s,c)$ and

$$p^\mu_{s,c} = \begin{cases} p_{s,c} & \text{if } g(s,c) < |\mathcal{G}| \\ p_{s,c} - \epsilon & \text{if } g(s,c) = |\mathcal{G}| \text{ and } z^\mu(s,c) = 1 \\ \min\left\{p_{s',c} \,:\, s' \in f(s) \text{ with } z^\mu(s',c) = 1 \text{ or } s' = s\right\} & \text{if } g(s,c) = |\mathcal{G}| \text{ and } z^\mu(s,c) = 0, \end{cases} \tag{2}$$

where ϵ is sufficiently small so the contingent tie-breaker of an effective priority provider does not break the initial order. Moreover, if two or more siblings receive the same contingent tie-breaker at school c due to an effective priority provider s, we resolve ties by assigning different contingent tie-breakers within the interval $(p_{s,c} - \epsilon, p_{s,c}]$, ensuring that their initial order is preserved.[8]

In both definitions, students who initially belong to a higher priority group (i.e., $g(s,c) < |\mathcal{G}|$) retain their group and random tie-breaker. Consequently, students from these higher priority groups do not gain additional benefits from contingent priorities, as it is common in practice.[9] However,

[8] For instance, $0 < \epsilon < \min\left\{\left|p_{s,c} - p_{a,c'}\right| : (s,c),(a,c') \in \mathcal{V}\right\}$. Then, if $s', s'' \in f(s) \setminus \{s\}$ and $p_{s',c} < p_{s'',c}$, we set $p^\mu_{s',c} = p_{s,c} - \epsilon'$ and $p^\mu_{s'',c} = p_{s,c} - \epsilon''$ such that $\epsilon' > \epsilon''$ and $\epsilon', \epsilon'' \in [0, \epsilon)$. Same logic applies for three or more siblings.

[9] In most school choice systems, students who belong to multiple groups (e.g., walk-zone and siblings priority) are not additionally prioritized and are treated as belonging to the higher priority group only.

these two definitions differ in their treatment of priority providers, receivers, and students without priority, capturing two "extremes". Under absolute priorities, groups are updated while maintaining the initial lotteries, so that students who provide or receive contingent priority (and who have initially no priority, i.e., $g(s,c) = |\mathcal{G}|$) stay in their initial group, while students without siblings or initial priority (i.e., $g(s,c) = |\mathcal{G}|$) move to a new group with the lowest priority (i.e., $g^{\mu}(s,c) = |\mathcal{G}|+1$). Note that effective providers of priority only get prioritized if they have a sibling assigned to the school; otherwise, they are considered as a single student, and thus, they have no priority. Conversely, under partial priorities, groups remain unchanged while lotteries are updated so that priority recipients inherit the (slightly perturbed) initial lottery of their effective priority provider (if beneficial), and students without priority retain their initial lotteries. As a result, a contingently prioritized student can displace any other student without priority under absolute priorities (as in the Chilean school choice system), while they can only displace students in their group with a worse tie-breaker than theirs or their effective provider (if any) under partial priorities.

We emphasize that, under our notions of contingent priorities, students granted such priority cannot displace applicants from groups with higher priority according to the initial group mapping g (e.g., students with *secured enrollment* or with priorities coming from enrolled siblings, as we discuss in Appendix E.1). This restriction can be readily adjusted if the clearinghouse wishes to permit students with contingent priority to displace those from these other groups. Finally, observe that the updated groups and lotteries resulting from a matching μ under both types of contingent priority induce a unique strict order $\succ_c^{\mu}$ for each school $c \in C$. As a result, we can use the contingent priority orders to define the concepts of *contingent justified-envy* and *contingent stability*.

DEFINITION 5 (CONTINGENT JUSTIFIED-ENVY AND STABILITY). Given a matching μ, a student s has *contingent justified-envy* towards another student s' assigned to school c if (i) $\ell(s) = \ell(s')$, (ii) $c \succ_s \mu(s)$, and (iii) $s \succ_c^{\mu} s'$. A matching is *contingent stable* if it is non-wasteful and if no student has *contingent justified-envy*.

REMARK 1. Our definition of contingent stability (in particular, the absolute case) significantly broadens the set of stable matchings to capture more complex scenarios involving multiple levels or sibling applicants. In Appendix A.1, we provide a series of instances illustrating the differences between contingent stability and other notions of stability in the literature of matching with couples.

3.3. Properties

In this section, we briefly discuss some properties of the underlying mechanisms to find stable matchings under contingent priorities. We defer the formal statements and proofs to Appendix B.

Existence and Complexity. The existence of a stable matching under contingent priorities and the complexity of finding one heavily depend on the type of priority and the tie-breaking rule. As we show in Proposition 3 and Theorem 3, a contingent stable matching under absolute priorities may not exist and the decision problem of certifying the (un)existence is NP-complete. Furthermore, as we show in Proposition 4 and Theorem 4, the same result holds for partial priorities under tie-breakers at the individual level. In contrast, if tie-breakers are at the family level, we show that partial contingent stability is equivalent to initial stability and, consequently, a stable matching always exists and can be found in polynomial time using the Deferred Acceptance algorithm.

Student-Optimality. As Example 4 in Appendix A.2 illustrates, the uniqueness of a student-optimal stable matching and the Rural Hospital Theorem do not hold under contingent priorities. Consequently, throughout this paper, we assume that the clearinghouse aims to find a stable matching under contingent priorities that is *rank-optimal*, i.e., one that minimizes the sum of assigned ranks across students' preference lists, assuming that being unassigned is preferable to any school not listed in their preferences.[10] Note that under the standard notion of stability, a rank-optimal stable matching coincides with the student-optimal stable matching (see e.g., Bobbio et al. (2022)).

Incentives. As we show in Propositions 6 and 7, the mechanisms to find rank-optimal contingent stable matchings are not strategy-proof unless we consider the partial case under family lotteries. Nevertheless, in Proposition 8, we show that these mechanisms are strategy-proof in the large (see Azevedo and Budish (2018)), i.e., it is approximately optimal for students to report their true preferences for any i.i.d. distribution of students' reports. Hence, in large markets such as the ones motivating this work, the lack of strategy-proofness is not a major concern.

Hard vs. Soft Priorities. Some of the "negative" results discussed in this section arise because the clearinghouse is compelled to grant contingent priorities to every student with an effective priority provider. However, the clearinghouse could have greater flexibility in deciding which priority providers to consider and which to disregard—similar to the handling of hard and soft quotas (Kojima 2012, Ehlers et al. 2014)—broadening the space of feasible stable matchings.

In the remainder of the paper, we say that contingent priorities are *hard* if every student with an effective priority provider must be prioritized according to Definitions 3 and 4 for the absolute and partial cases, respectively. On the other hand, we say that contingent priorities are *soft* if the clearinghouse can ignore some effective priority providers and treat their siblings as students

[10] Under this objective, a lower average rank indicates that students are matched to more desirable schools.

without priority. Formally, whenever a pair $(s, c) \in \mathcal{V}$ satisfies the conditions in Definition 2 for a given matching μ, hard priorities imply that $z^{\mu}(s, c) = 1$, while soft priorities imply that $z^{\mu}(s, c) \in \{0, 1\}$, with the clearinghouse being able to choose either depending on its goals. For example, if the clearinghouse sets $z^{\mu}(s, c) = 0$ for every effective priority provider, then the clearinghouse can entirely ignore contingent priorities and target an initial stable matching. Finally, we say that contingent priorities are *hybrid* if the clearinghouse enforces consideration of only a subset of effective priority providers, striking a balance between hard and soft policies. Such hybrid approach can be viewed as points along a continuum of "softness," ranging from hard (all providers enforced) to soft (none enforced). For example, the mechanism may impose lower bound requirements on how many providers must be honored, as numerically explored in Section 5.3.1, or introduce similar rules that guarantee selective—but not universal—enforcement of contingent priorities.[11]

Ultimately, the choice between hard, soft or hybrid contingent priorities depends on the policy-makers' goals and their capacity to discriminate among students. Hard priorities may be desirable if the clearinghouse must treat all students equally, whereas soft or hybrid priorities me be preferable when ensuring the existence of a stable matching is the primary concern.

Transparency. A growing concern in school choice systems is the limited information available to students for understanding how assignments are determined and whether they are correct (Abdulkadiroglu et al. 2017). Hakimov and Raghavan (2022) highlight two distinct challenges: (i) the Verifiability Problem, where students may doubt the correctness of their individual assignment, and (ii) the Transparency Problem, where students may question whether the assignment process was implemented as promised.

Our proposed framework addresses both concerns. Since all necessary inputs—students' preferences, school priorities, and capacities—as well as the assignment algorithm itself are publicly available, the resulting matchings are both transparent and verifiable. Moreover, even if the input data were not public, our mechanisms would still be verifiable, as the assignment can be fully recovered considering (i) the combination of group and lottery values that result from the contingent priority orders as eligibility criteria; and (ii) the minimum group and lottery among the students admitted to each level in each school as cutoffs. Specifically, given the matching μ, the eligibility criteria for student s at school c would be $\rho^{\mu}_{s,c} = \left(g^{\mu}_{s,c}, p^{\mu}_{s,c}\right)$, and the cutoff for school c in $\ell(s)$ would

[11] For instance, hybrid rules could be implemented by (i) modifying the objective function, such as introducing penalties for ignoring providers or assigning extra weight to certain families (e.g., larger families or those with siblings applying to the same level), or (ii) adding constraints, such as school-level caps on how many providers can be disregarded or system-wide thresholds requiring that at least a fixed proportion of effective providers receive priority.

be $\bar{\rho}^{\mu}_{c,\ell(s)} = \left(\bar{g}^{\mu}_{c,\ell(s)}, \bar{p}^{\mu}_{c,\ell(s)}\right)$, where $\bar{g}^{\mu}_{c,\ell(s)} = \max\left\{g \in \mathcal{G} : \exists s' \in \mathcal{S}^{\ell(s)},\ \mu(s') = c,\ g(s',c) = g\right\}$ and $\bar{p}^{\mu}_{c,\ell(s)} = \max\left\{p \in \mathbb{R}_+ : \exists s' \in \mathcal{S}^{\ell(s)},\ \mu(s') = c,\ p_{s',c} = p\right\}$. Then, students can *verify* their assignment by checking that (i) $\rho^{\mu}_{s,c'} > \bar{\rho}^{\mu}_{c,\ell(s)}$ for all $c' \succ_s \mu(s)$ and (ii) $\rho^{\mu}_{s,\mu(s)} \leq \bar{\rho}^{\mu}_{\mu(s),\ell(s)}$, where the comparison of pairs of eligibility criteria is done lexicographically.

4. Formulations

The results in Section 3.3 imply that there is no hope of designing a polynomial-time approach to finding a stable matching with contingent priorities, unless NP = P. This motivates our use of integer linear programming to obtain the rank-optimal stable matchings for each type of contingent priority (if it exists), which is the focus of this section.

The formulations we present in Sections 4.1 and 4.2 (for absolute and partial, respectively) extend that in (Baïou and Balinski 2000) to find a rank-optimal stable matching that accounts for contingent priorities. Specifically, let $r_{s,c}$ be the position of school $c \in \mathcal{C}$ in student s's preference list, and let $r_{s,\emptyset}$ be a parameter that captures the cost of having student s unassigned. Then, a rank-optimal (initial) stable matching corresponds to the solution of the following integer program:

$$\min \sum_{(s,c)\in\mathcal{V}} r_{s,c} \cdot x_{s,c}$$

$$s.t.\ q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c'\in\mathcal{C}: \\ c' \succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{s'\in\mathcal{S}^{\ell(s)}: \\ s' \succ_c s}} x_{s',c}, \qquad \forall (s,c) \in \mathcal{S}\times\mathcal{C}, \tag{3a}$$

$$\mathbf{x} \in \mathcal{P}, \tag{3b}$$

where

$$\mathcal{P} = \left\{\mathbf{x} \in \{0,1\}^{\mathcal{V}} : \sum_{\substack{c\in\mathcal{C}\cup\{\emptyset\}: \\ (s,c)\in\mathcal{V}}} x_{s,c} = 1,\ \forall\, s \in \mathcal{S}, \quad \sum_{\substack{s\in\mathcal{S}^{\ell}: \\ (s,c)\in\mathcal{V}}} x_{s,c} \leq q_c^{\ell},\ \forall\, c \in \mathcal{C},\ \ell \in \mathcal{L}\right\}$$

is the set of feasible assignments, i.e., $\mathbf{x}$ encodes the matching of students to schools and $\mathbf{x} \in \mathcal{P}$ ensures that each student is assigned to at most one school and that each school does not exceed its capacity in each level. The objective is to minimize the preference of assignment of each student, and the set of constraints (3a) guarantees that student s has no initial justified-envy in school c.

To incorporate contingent priorities, we extend Formulation (3) as follows. With a slight abuse of notation, let $f(s,c) = \{s' \in f(s)\ :\ (s',c) \in \mathcal{V}\}$, i.e., $f(s,c)$ captures the family members of $f(s)$ who apply to school c. Then, we define the set of variables $z_{s,c} \in \{0,1\}$ for all $(s,c) \in \mathcal{V}$ with

$|f(s,c)| \geq 2$ and $c \in C$ (i.e., student s has siblings also participating in the admissions process and who prefer c over being unassigned), where $z_{s,c}$ is equal to one if student s is an effective priority provider in school c and one of their siblings is also assigned to c, and zero otherwise. Given a set of decision variables $\mathbf{x}$ (i.e., a matching), the set of variables $\mathbf{z}$ that are consistent with the model discussed in Section 3 can be characterized as follows:

$$\mathcal{R}(\mathbf{x}) = \Bigg\{ \mathbf{z} \in \{0,1\}^{\mathcal{V}} : \; z_{s,c} \leq x_{s,c}, \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; c \in C \tag{4a}$$

$$z_{s,c} \leq \sum_{s' \in f(s) \setminus \{s\}} x_{s',c}, \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; c \in C \tag{4b}$$

$$\sum_{\substack{s' \in \mathcal{S}^{\ell(s)} \\ s' \succ_c s}} \sum_{\substack{c' \in C \cup \{\emptyset\} \\ c \succeq_{s'} c'}} x_{s',c'} \leq \left(q_c^{\ell(s)} - 1\right) + \left|\mathcal{S}^{\ell(s)}\right| \cdot (1 - z_{s,c}), \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; c \in C \tag{4c}$$

$$\sum_{s \in f} z_{s,c} \leq 1, \qquad \forall\, f \in \mathcal{F} : |f(s,c)| \geq 2,\; c \in C \Bigg\}. \tag{4d}$$

Following Definition 2 and the definition of $\mathbf{z}^{\mu}$ given in Section 3.2, s provides priority to their siblings in school c if (i) they get matched to it, captured by the set of constraints (4a); (ii) at least one of their siblings potentially benefits from it, captured by the set of constraints (4b); and (iii) there are at most $q_c^{\ell(s)} - 1$ students who initially have higher priority in c and weakly prefer c over their matching, captured by the set of constraints (4c). Finally, the set of constraints (4d) ensures that at most one member of each family provides contingent priority to their siblings in each school.

To capture the action of receiving contingent priority and benefiting from it, we consider the set of variables $y_{s,s',c} \in \{0,1\}$ for all $s \in \mathcal{S}$ with $|f(s,c)| \geq 2$, and $s' \in f(s) \setminus \{s\}$, where $y_{s,s',c}$ is equal to one if student s' gets matched to c having s as their effective priority provider, and zero otherwise. Thus, given a set of decision variables $\mathbf{x} \in \{0,1\}^{\mathcal{V}}$ and $\mathbf{z} \in \{0,1\}^{\mathcal{V}}$, the set of variables $\mathbf{y}$ can be fully characterized as follows:

$$\mathcal{Q}(\mathbf{x},\mathbf{z}) = \Bigg\{ \mathbf{y} \in \{0,1\}^{\mathcal{S} \times \mathcal{V}} : y_{s,s',c} \leq x_{s',c}, \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; s' \in f(s),\; c \in C \tag{5a}$$

$$y_{s,s',c} \leq z_{s,c}, \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; s' \in f(s),\; c \in C \tag{5b}$$

$$y_{s',s,c} \leq 1 - z_{s,c}, \qquad \forall\, s \in \mathcal{S} : |f(s,c)| \geq 2,\; s' \in f(s),\; c \in C \Bigg\}. \tag{5c}$$

The sets of constraints (5a) and (5b) guarantee that student s' can receive contingent priority from s in school c only if the former gets assigned and the latter is their family's provider of priority in that school, respectively. Finally, the set of constraints (5c) ensures that students do not simultaneously provide and receive contingent priority.

REMARK 2. To ease exposition, in the remainder of the paper, we assume that $|\mathcal{G}| = 1$, i.e., there is initially a single group and, thus, the initial order in each school c, $\succ_c$, is solely determined by the initial random tie-breakers. In Appendix E, we discuss how to incorporate other groups.

REMARK 3. We omit constraints that explicitly require $z_{s,c} = 1$ for the student s with the most favorable lottery among the family members in $f(s)$ satisfying Definition 2 since such constraints are unnecessary in both the absolute and partial cases, as there always exists a feasible solution leading to the same allocation that inherently satisfies this condition (see Lemma 1 in Appendix C.1). More precisely, given a feasible solution, we can construct another feasible solution with the same matching, i.e., variables $\mathbf{x}$, and modified variables $\mathbf{z}$ (and $\mathbf{y}$) to respect the condition.

4.1. Absolute Priority

As discussed in Definition 3, a student s who either provides or receives absolute contingent priority can displace any other student who does not hold such priority, regardless of their random tie-breakers. However, when two students belong to the same priority group, they are ranked according to their tie-breakers. Consequently, we can formulate the problem of finding a rank-optimal stable matching with absolute contingent priority as follows:

$$\min \quad \sum_{(s,c)\in\mathcal{V}} r_{s,c}\cdot x_{s,c}$$

$$s.t. \quad q_c^{\ell(s)}\cdot\left(1-\sum_{\substack{c'\in C:\\ c'\succeq_s c}} x_{s,c'}\right) \le \sum_{\substack{a\in\mathcal{S}^{\ell(s)}:\\ a\succ_c s}} x_{a,c} + \sum_{\substack{f\in\mathcal{F}:\\ |f|\ge 2}}\ \sum_{\substack{\{a,a'\}\subseteq f:\\ a\in\mathcal{S}^{\ell(s)},\\ a\prec_c s}} y_{a',a,c} + \sum_{\substack{a\in\mathcal{S}^{\ell(s)}:\\ a\prec_c s}} z_{a,c}, \quad \forall (s,c)\in\mathcal{V}, \tag{6a}$$

$$x_{s',c} + \left(1-\sum_{\substack{c'\in C:\\ c'\succeq_s c}} x_{s,c'}\right) \le 2 - x_{a,c} + 1_{\{a\succ_c s\}}\cdot\left(z_{a,c} + \sum_{a'\in f(a)\setminus\{a\}} y_{a',a,c}\right),$$
$$\forall c\in C, f\in\mathcal{F}, \{s,s'\}\subseteq f, a\in\mathcal{S}^{\ell(s)}\setminus f, \tag{6b}$$

$$\mathbf{x}\in\mathcal{P},\ \mathbf{z}\in\mathcal{R}(\mathbf{x}),\ \mathbf{y}\in\mathcal{Q}(\mathbf{x},\mathbf{z}). \tag{6c}$$

The first set of constraints (6a) extends (3a) to incorporate absolute contingent priorities. Specifically, suppose that student s is not matched with school c or better. This set of constraints implies that there are at least $q_c^{\ell(s)}$ students matched to school c at level $\ell(s)$ who are either (i) initially more preferred than student s (first term on the right-hand side), (ii) initially less preferred than s but receive contingent priority from one of their siblings (second term in the right-hand side), or (iii) initially less preferred than s but provide contingent priority to their siblings (third term in the right-hand side). The latter two terms capture that, according to Definition 3, receivers and providers of priority keep their initial group, while students with no contingent priority move to the new (lowest) priority group $|\mathcal{G}|+1$ and, thus, are less preferred under the contingent matching $\mathbf{x}$.[12] The second set of constraints (6b) ensures that ties among prioritized students are broken according to their initial random tie-breakers (or, equivalently, using the initial priority order due to Remark 2). Namely, if student s has a sibling s' matched to c (potentially in a different level) and s is not matched to c or better, then no other student $a \in \mathcal{S}^{\ell(s)} \setminus f(s)$ can get matched to c unless their random-tie breaker is better than that of s and either they receive or effectively provide priority. Moreover, this set of constraints ensures that prioritized students are never displaced by students without priority. To see this, note that if a has no siblings, the sum on the right-hand side is zero and, consequently, if s' is assigned to c and s is not assigned to c or better, then $x_{a,c}$ is forced to be zero. In Theorem 1, we formally show that the feasible region (21) correctly captures the notion of contingent absolute priorities. We defer the proof of this result to Appendix C.2.

THEOREM 1. *Feasible region* (21) *is a valid formulation of the set of contingent stable matchings under absolute priorities.*

4.1.1. Hard vs. Soft Absolute Contingent Priorities. We now explore how to incorporate hard and soft absolute priorities in Formulation (21). As discussed in Section 3.3, these priorities are captured by the way we assign values to $z^{\mu}(s,c) \in \{0,1\}$ for each pair $(s,c) \in \mathcal{V}$ under a given matching μ. Recall that, regardless of whether priorities are hard or soft, we assign $z^{\mu}(s,c) = 0$ for every student s such that $f(s) = \{s\}$. Hence, the differences between hard and soft arise when we consider students with siblings.

Hard Priorities. In this case, $z^{\mu}(s,c) = 1$ whenever s is an effective priority provider in school c, i.e., s has the most favorable random tie-breaker among their family members satisfying the conditions in Definition 2 at school c. In Theorem 1, we show that the set of constraints (21) ensures that $z_{s,c} = 1$ for all such pairs satisfying $|f(s) \cap \mu(c)| \geq 2$ (if a feasible solution exists). This result relies on Lemma 2 (see Appendix C), which guarantees that whenever a family of size at least two has one of their members matched to a school and another to a less preferred one, then it *must* have an effective priority provider in that school. Key to ensure this result and enforce absolute priorities

[12] Recall that $z_{a,c} = 1$ implies that student a can rightfully claim a seat at school c by 4c, allowing them to retain their initial priority group and thus outrank s, who moves to the new, lower priority group $|\mathcal{G}|+1$.

is the set of constraints (6b). To see this, suppose there is a family $f = \{s, s'\}$ where $x_{s',c} = 1$ and $x_{s,c'} = 0$ for all $c' \succeq_s c$, so that the left-hand side is equal to two. Then, (6b) can be re-written as

$$x_{a,c} \leq 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \right), \quad \forall a \in \mathcal{S}^{\ell(s)} \setminus f.$$

Note that if a is such that $a \prec_c s$, then the right-hand side is zero, forcing $x_{a,c} = 0$ and granting that s has no contingent justified envy towards a. Conversely, we could have $x_{a,c} = 1$ if a either provides or receives priority, ensuring that the family $f(a)$ has an effective provider of priority and guaranteeing that s has no contingent justified envy (as $a \succ_c s$).

Soft Priorities. In this case, the clearinghouse can ignore some effective priority providers and set $z^\mu(s,c) = 0$ depending on its goals. To capture this, we relax the set of constraints (6b) as follows

$$z_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'} \right) \leq 2 - x_{a,c} + 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \right). \tag{7d}$$

Note that the only change comes from replacing $x_{s',c}$ with $z_{s',c}$ on the left-hand side, which means that having a sibling s' matched to a school c is not enough for student s to displace another student $a \notin f(s)$, i.e., the discussion for hard absolute priorities above does not apply here. Instead, the displacement will occur if and only if $z_{s',c} = 1$. In Proposition 1, we show that the set of constraints (7d) allows us to capture soft absolute priorities and, furthermore, an entire spectrum of *hybrid* solutions ranging from the initially stable matchings to those with hard absolute priorities. We defer the proof to the Appendix C.3.3.

PROPOSITION 1. *For every point* $(\mathbf{x}, \mathbf{z}, \mathbf{y})$ *feasible for Formulation* (21)*, we can construct* $(\mathbf{x}', \mathbf{z}', \mathbf{y}')$ *feasible for Formulation* (21) *with* (6b) *replaced by* (7d)*. Moreover, this modified formulation is always feasible as it contains the set of initially stable matchings.*

4.2. Partial Priority

The key difference between absolute and partial is that, in the former case, a prioritized student can move any other student without siblings matched to the school. In the latter, in contrast, prioritized students can only take over those who have no siblings assigned to the school if their contingent tie-breaker (the most favorable between theirs and that of their effective priority provider) is better.

To capture this, we modify the set of constraints in (21) in three important ways. First, we remove the terms involving $z_{a,c}$ from both (6a) and (6b) since effective priority providers keep their initial group under partial priorities. Second, we modify the set of constraints (6a) by adding the condition $a' \succ_c s$ in the second summation of the right-hand side to ensure that the student a' providing the siblings' priority has a better tie-breaker than the student s who gets displaced by their prioritized sibling. Finally, we modify the set of constraints (6b) by multiplying $x_{a,c}$ and $y_{a',a,c}$ by the indicators $1_{\{a \prec_c s \wedge s'\}}$ and $1_{\{a \wedge a' \succ_c s \wedge s'\}}$, respectively, where $s \wedge s'$ denotes the best initial priority order among the students s, s'.[13] To see the role played by these indicators, let $f = \{s, s'\}$ be a family with $x_{s',c=1}$ and $x_{s,c'} = 0$ for all $c' \succeq_s c$, and consider an additional student $a \in \mathcal{S}^{\ell(s)} \setminus f$. Then,

- If $s \succ_c a$, then the constraint ensures that a can only be matched to c (i.e., $x_{a,c} = 1$) if they receive contingent priority from a sibling a' (i.e., $y_{a',a,c} = 1$) with higher initial priority than both s and s' (i.e., $1_{\{a' \wedge a \succ_c s' \wedge s\}} = 1_{\{a' \succ_c s' \wedge s\}} = 1$). Otherwise, the constraint forces $x_{a,c} = 0$.
- If $s' \succ_c a \succ_c s$, then the constraint ensures that a can only be matched to c if they receive contingent priority from a sibling a' with higher initial priority than both s and s'. Otherwise, the terms $y_{a',a,c}$ are equal to zero, which forces $x_{a,c} = 0$.

As a result, we formulate the problem of finding a stable assignment with partial priorities as:

$$
\begin{aligned}
\min \quad & \sum_{(s,c)\in\mathcal{V}} r_{s,c} \cdot x_{s,c} \\
s.t. \quad & q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)} \\ a \prec_c s \prec_c a'}} y_{a',a,c}, \qquad \forall (s,c) \in \mathcal{V}, && \text{(7a)} \\
& x_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq 2 - x_{a,c} \cdot 1_{\{a \prec_c s \wedge s'\}} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}}, \\
& \qquad \forall c \in C, f \in \mathcal{F}, \{s,s'\} \subseteq f, a \in \mathcal{S}^{\ell(s)} \setminus f, && \text{(7b)} \\
& \mathbf{x} \in \mathcal{P},\ \mathbf{z} \in \mathcal{R}(\mathbf{x}),\ \mathbf{y} \in \mathcal{Q}(\mathbf{x},\mathbf{z}). && \text{(7c)}
\end{aligned}
$$

In Theorem 2, we show that the feasible region (7) correctly captures the notion of contingent partial priorities. We defer the proof of this result to Appendix C.3.

THEOREM 2. *The feasible region* (7) *is a valid formulation of the set of contingent stable matchings under partial priorities.*

4.2.1. Hard vs. Soft Partial Contingent Priorities. On the one hand, Formulation (7) guarantees that the optimal solution adheres to hard partial priorities. We omit the formal proof here,

[13] Concretely, $s \wedge s' \succ_c a$ if and only if $s \succ_c a$ or $s' \succ_c a$, and $s \wedge s' \succ_c a \wedge a'$ if and only if $s \wedge s' \succ_c a$ and $s \wedge s' \succ_c a'$.

as it parallels the reasoning used in the absolute case. On the other hand, to account for soft partial priorities, we replace constraint (7b) with the following set of constraints:

$$z_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq 2 - x_{a,c} \cdot 1_{\{a \prec_c s \wedge s'\}} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}}. \tag{8d}$$

As in the absolute case, note that the only difference between (7b) and (8d) is the use of $z_{s',c}$ instead of $x_{s',c}$. In Proposition 2, we show that this change guarantees that the problem is always feasible and that we can recover a range of stable matchings with partial priorities varying $\mathbf{z}$.

PROPOSITION 2. *For every point* $(\mathbf{x}, \mathbf{z}, \mathbf{y})$ *feasible for Formulation* (7)*, we can construct* $(\mathbf{x}', \mathbf{z}', \mathbf{y}')$ *feasible for Formulation* (7) *with* (7b) *replaced* (8d)*. Moreover, this modified formulation is always feasible as it contains the set of initially stable matchings.*

5. Application to School Choice in Chile

We have collaborated closely with the Chilean Ministry of Education and Consilium Bots—the NGO responsible for implementing the centralized school assignment process—to evaluate our framework for accommodating contingent priorities, which ultimately led to its inclusion in the Ministry's official list of prospective improvements to the assignment system.[14] In what follows, we present an empirical evaluation of our framework using administrative data from the Chilean school choice system, illustrating its potential impact on assignment outcomes and highlighting the value of incorporating contingent priorities in practice.

5.1. Background

The Chilean school choice system—introduced in 2016 and fully implemented in 2020—governs admissions to all schools that receive public funding, covering every grade level from Pre-K through 12th grade and currently serving more than half a million students each year. Under this centralized system, families report strict preference lists for each of their children,[15] and schools order applicants according to five legislated priority groups: (i) students with siblings enrolled for the next academic year; (ii) students with older siblings simultaneously participating in the admissions process and assigned to that school; (iii) children of school employees; (iv) returning

[14] For reasons unrelated to the technical merits of our proposal, it was not deployed in the most recent policy cycle.

[15] To our knowledge, no centralized school choice system allows families to submit joint preference lists over assignments of multiple children, i.e., to report preferences over combinations of schools for siblings rather than individual rankings for each child (see Correa et al. (2022)). Hence, our focus on individual preference reports per child is without loss of practical relevance.

former students; (v) all remaining applicants. Priorities are processed in this strict order: priority from siblings enrolled always takes precedence, followed by priority from older siblings assigned to the school, and then the remaining groups in the order listed above.[16] Ties within each priority group are resolved using a multiple tie-breaking rule applied at the family level, whereby each family receives an independent random tie-breaker for every school, slightly perturbed to break ties among siblings while preserving the relative order between families.

5.2. Simulation Setting

5.2.1. Data. We use data from the admissions process in 2023 and consider all students who applied to the system in the southernmost region of the country (Magallanes).[17] We focus on this region for three reasons: (i) it is the pilot region where new policy changes are first evaluated; (ii) its geographical isolation ensures that applicants only list local schools, avoiding cross-region interactions; and (iii) the composition of students and schools is representative of the rest of the country, while the size of the instance allows us to speed up computations. Appendix D reports descriptive statistics comparing regions and presents our main results for three additional ones (Atacama, O'Higgins and Los Lagos),[18] confirming the robustness of our findings.

5.2.2. Benchmarks. We compare our framework with the algorithm currently used to solve the Chilean school choice problem, which we refer to as *Descending*. After collecting families' preferences and sorting students at each school, the clearinghouse runs an algorithm that processes levels sequentially in decreasing order, i.e., from the highest (12th grade) to the lowest (Pre-K). For each level ℓ, the algorithm:

1. Updates schools' priorities to account for students who may benefit from having an older sibling previously assigned to the same school.
2. Runs the student-proposing Deferred Acceptance algorithm using the updated preferences and priorities for students and schools in level ℓ.

Note that this algorithm limits siblings' priorities to be *one-directional*: only older students can be effective priority providers for their younger siblings. Consequently, this approach does not lead to

[16] Students with no siblings enrolled for the next year, with only younger siblings or with no siblings at all fall into groups (iii)–(v). As a result, they can be displaced by students benefiting from (i) or (ii), but never the other way around. In Appendix E, we extend our formulations to account for (i).

[17] We use actual families, students' reported preferences, and schools' capacities. All data is publicly available on this website.

[18] We chose these regions to provide geographic coverage (north, center, and south, respectively), span different market sizes (10k–30k applicants), and reflect the national share of students with siblings applying concurrently.

a stable matching nor is strategy-proof in general (Correa et al. 2022), and it may lead to matchings that are neither stable under contingent priorities nor rank-optimal (see Appendix D.1.1).[19]

Finally, we also compare our approaches with (i) the student-optimal stable matching (SOSM) obtained by the Deferred Acceptance algorithm (i.e., assuming no one can benefit from having contingent priorities) and (ii) the *family-oriented stable matching* (FOSM), which corresponds to the standard stable matching that maximizes the number of family members assigned to the same school; we include the integer linear programming formulation to obtain FOSM in Appendix D.1.3.

5.2.3. Setup. For each benchmark, we run 100 simulations, each using a different draw of the random tie-breakers to solve the instance based on the resulting priorities and students' reported preferences. Consistent with the current practice in Chile, we report results under the multiple tie-breaking rule at the family level (MTB-F) and defer results for alternative tie-breaking rules to Appendix D.3. To simplify the analysis and exposition, we assume that all students initially belong to the same priority group (i.e., there is a single initial priority group). This is without major loss of practical relevance, as sibling priorities have prevalence over all other priority groups; Appendix E discusses how the framework can be extended to incorporate heterogeneous priority structures. Finally, all mathematical programming formulations are solved using Gurobi 11.0, with a MIPGap tolerance of 0.1% and no time limit.[20] In the formulations for absolute priorities (21), partial priorities (7), and FOSM (18), we set the penalty for unassigned students to $r_{s,\emptyset} = |\succ_s| + 1$.[21]

5.3. Results

In Table 1, we report a summary of our main simulation results. The first two columns show the number of instances successfully solved (out of 100) and the average preference of assignment, respectively. Note that the latter plus the penalty for being unassigned captures the objective function being optimized, as discussed in Section 4. The next two pairs of columns (under *Top Pref.* and *Unassigned*) report the average and standard error for the number of students assigned to their top preference and those left unassigned, respectively. The *Together* columns present the average and standard error for the number of students matched to the same school as at least one of their

[19] In Appendix D.1.1, we provide a mathematical programming formulation that modifies Absolute-Hard by restricting the direction of sibling priorities and preventing providers from being prioritized to compute the best possible "descending" stable matching.

[20] Average running times for Absolute-Hard and Absolute-Soft are 10.18 and 15.75 seconds, respectively, with standard errors of 0.86 and 1.32 seconds. The corresponding average MIPGaps are 2.37×10^{-4} and 7.49×10^{-4}. See `www.gurobi.com` for details.

[21] Results are qualitatively similar if we instead set $r_{s,\emptyset} = |C| + 1$, i.e., imposing a large penalty for unassignment. As discussed in (Bobbio et al. 2022), larger penalties reduce the number of unassigned students, whereas smaller penalties improve the rank of assignments for more students.

Table 1 Comparison of Benchmarks

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Hard	86	1.910	2193.77	0.81	937.94	1.04	664.43	0.33	203.66	0.83	147.52	0.66	153.78	0.56
Soft	100	1.899	2196.49	1.00	941.73	1.02	569.39	0.95	200.41	1.29	201.06	1.55	208.32	1.06
FOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
SOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
Descending	100	1.910	2192.33	0.95	943.80	1.03	556.84	1.78	200.95	1.30	202.88	1.17	224.69	1.29

Table 2 Sensitivity to Softness

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Hard	86	1.910	2193.77	0.81	937.94	1.04	664.43	0.33	203.66	0.83	147.52	0.66	153.78	0.56
Soft	100	1.899	2196.49	1.00	941.73	1.02	569.39	0.95	200.41	1.29	201.06	1.55	208.32	1.06
FOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
SOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
Descending	100	1.910	2192.33	0.95	943.80	1.03	556.84	1.78	200.95	1.30	202.88	1.17	224.69	1.29
Simultaneous	100	1.917	2203.30	1.00	901.71	1.32	665.48	0.44	194.61	1.23	147.48	0.92	161.28	0.39

siblings. The *Separated* columns examine cases where siblings who applied to at least one school in common were assigned to different schools. The *None* columns report the average and standard error for the number of students whose siblings, as well as themselves, were left unassigned. The *One* columns report the same statistics for students where only one sibling is matched, and at least one school on both siblings' lists is preferred over their assigned match. Finally, the *Both* columns present these statistics for students where both siblings are matched to different schools, despite sharing a school they both prefer.[22]

First, we find that the problem of finding a stable assignment with hard absolute contingent priorities is feasible for roughly nine out of ten of the simulations considered, while it is always feasible in the soft case (as shown in Proposition 1).

[22] Note that, in the last three cases, we may double count in cases of families with more than two applicants. Nevertheless, only 69 out of 682 families with multiple applicants involve three or more students.

Second, we observe that the results for FOSM and SOSM are identical,[23] whereas Absolute-Soft and Descending yield similar numbers of siblings assigned together. However, Absolute-Soft offers a slight advantage in terms of average preference of assignment, the number of students receiving their top choice, and the number of unassigned students. Moreover, Table 8 in Appendix D.3 shows that *Ascending*—i.e., processing grades in ascending order starting from the lowest level—results in more siblings being placed together.

Third, among the solved instances, Absolute-Hard performs similarly to Descending in terms of assignment preferences (both average and number in top preference), while achieving a lower number of unassigned students. More importantly, Absolute-Hard substantially increases the number of siblings assigned together, yielding a 19.5% improvement (from 556.84 to 664.43).[24] The right-hand panel of Table 1 indicates that this improvement arises primarily from families where at most one of the siblings remains unassigned (columns One and Both). Intuitively, absolute contingent priorities enable students with unfavorable lottery numbers—who would otherwise likely remain unmatched or assigned to a lower preference—to leverage their sibling's priority—coming from their effective provider, who must rightfully claim a spot—, increasing their chances of admission to a more preferred school. These findings are robust to the choice of tie-breaking rule (see Appendix D.3) and likely represent a conservative estimate of the potential benefits. In fact, the gains in the number of siblings assigned together when moving from Descending to Absolute-Hard are even larger in other regions—24.48% in Atacama, 25.48% in O'Higgins, and 28.21% in Los Lagos (see Tables 11, 12 and 13 in Appendix D.4).

Finally, Table 14 in Appendix D.5 shows that increasing the number of siblings assigned together is accompanied by improvements in other outcomes for students with siblings: their average preference assignment decreases, the fraction placed in their top choice rises, and the share of unassigned students falls relative to Descending. These gains come with some costs for students without siblings, but the magnitude of these losses is smaller than the improvements for those with siblings. Importantly, under the current algorithm (Descending), students with siblings have worse outcomes than those without. Thus, applying contingent priorities not only yields larger absolute gains, but also helps to close an existing gap, enhancing fairness rather than undermining it.

[23] A likely explanation is that, in large markets, the core of stable matchings is typically small (Ashlagi et al. 2017), leaving few feasible assignments that improve family-level outcomes without violating stability.

[24] We obtain similar results if we restrict the comparison to those instances solved by Absolute-Hard.

Overall, these results suggest that absolute priorities effectively prioritize students with siblings and increase their joint assignments, while the standard notion of stability (i.e., initial stability) has a limited impact on keeping families together.

5.3.1. Sensitivity to "Softness". One limitation of soft priorities is that they must balance prioritizing students with siblings against potentially harming the objective function. For instance, by receiving contingent priority, a student may displace others and have a negative overall effect on the distribution of assignment preferences, leading to a higher objective function value. Consequently, relying solely on soft priorities may exclude too many effective priority providers in an attempt to minimize any negative impact on the objective function.

As discussed in Section 3.3, hybrid approaches allow for intermediate enforcement of contingent priorities, lying between the hard and soft extremes. To illustrate this idea and explore its potential benefits, we introduce a hybrid variant that imposes a minimum requirement on the number of effective providers considered by adding the following set of constraints:

$$\sum_{c\in C}\sum_{s\in\mathcal{S}} z_{s,c} \geq \zeta, \tag{8}$$

where ζ represents the minimum number of effective priority providers the solution must incorporate. By varying ζ, we can examine how the mechanism performs under different enforcement levels, highlighting the trade-off between increasing sibling assignments and preserving a favorable overall distribution of preferences, and showing how hybrid approaches allow for intermediate solutions between hard and soft priorities.

In Table 3, we report simulation results similar to those reported before,[25] but now adding the set of constraints (8) in the formulation of absolute-soft priorities (i.e., Formulation (21) with (6b) replaced by (7d)) and varying the value of ζ. For comparison, we also add the results of Absolute-Hard and Absolute-Soft (i.e., considering $\zeta = 0$). First, we observe that the problem is feasible for all instances considered when $\zeta < 320$, while the problem is always infeasible when $\zeta > 320$. Second, we observe a slight improvement on the average preference of assignment and in the number of unassigned student and a slight decrease in the number of students assigned to their top preference when increasing the number of siblings matched together (through increasing ζ). In contrast, the number of siblings who benefit from contingent priorities significantly increases as we increase ζ (see the Together column). For instance, when comparing the results with $\zeta = 0$ (Absolute-Soft)

[25] Specifically, we perform 100 simulations for each value of ζ considering the same simulation setup discussed above, and then we compute the average across these simulations for each outcome of interest.

Table 3 Sensitivity to Softness

			Top Pref.		Unassigned		Together		Separated: None		Separated: One		Separated: Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
280	100	1.900	2197.1	0.98	940.13	1.04	594.39	0.52	200.15	1.19	190.53	1.4	194.09	1.07
290	100	1.901	2195.27	0.9	939.13	0.95	608.7	0.28	200.31	1.22	183.69	1.4	185.93	1.02
300	100	1.902	2194.45	0.74	938.74	0.98	627.45	0.21	200.07	1.19	174.42	1.37	175.55	0.98
310	100	1.905	2193.65	0.78	937.18	0.92	646.93	0.19	200.29	1.2	162.99	1.4	166.17	0.93
320	80	1.908	2192.51	1.01	933.33	0.61	667.21	0.19	195.21	1.16	153.69	1.55	157.6	0.69
Hard	86	1.91	2193.77	0.81	937.94	1.04	664.43	0.33	203.66	0.83	147.52	0.66	153.78	0.56
Soft	100	1.899	2196.49	1.0	941.73	1.02	569.39	0.95	200.41	1.29	201.06	1.55	208.32	1.06

and $\zeta = 320$, we observe that the number of students matched with their siblings increases roughly by 17.18% (from 569.39 to 667.21). Third, we observe that the primary source of improvement in the joint assignment of siblings comes from reducing the number of cases where both siblings are assigned to different schools (see the Both column), which drops by more than 30% (from 154.66 when $\zeta = 0$ to 108.35 when $\zeta = 300$). Finally, comparing the results in Table 3 for $\zeta = 310$ with those for Descending (in Table 1), we observe that the former dominates the latter in every dimension, as the number of students assigned to their top preference is higher, those unassigned is lower, and more siblings are assigned together.

These results suggest that a hybrid approach that combines a soft implementation of contingent priorities with constraints that enforce the solution to incorporate a minimum number of effective priority providers achieves a good balance between feasibility, benefiting students with siblings, and not harming the distribution of preference of assignment.

5.3.2. Policy Alternatives to Maximize Joint Assignments. Although Chilean law mandates that the final matching must be rank-optimal, stable, and that ties must be broken using a multiple tie-breaking rule at the family level, relaxing some of these requirements may help us understand the potential benefits of alternative policy designs that our framework can accommodate. To explore this, we evaluate two modifications to the current rules: (i) changing the objective function from minimizing the sum of ranks of assignments to maximizing the number of receivers of sibling priority, i.e., maximizing $\sum y_{s,s',c}$;[26] and (ii) removing initial random tie-breakers, allowing our

[26] We also considered other objectives aimed at favoring the joint assignment of siblings, such as maximizing the number of priority providers or the number of siblings assigned together (as in FOSM). Results were qualitatively similar.

Table 4 Policy Alternatives

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Panel 1: Changing Objective														
280	100	1.922	2182.31	0.65	918.97	1.12	625.20	0.93	197.25	1.31	155.62	1.26	197.91	0.91
290	100	1.922	2182.44	0.67	918.82	1.11	625.83	0.73	197.01	1.32	155.50	1.20	197.94	0.86
300	100	1.922	2182.70	0.69	919.73	1.08	631.01	0.37	197.29	1.32	154.39	1.19	192.78	0.75
310	100	1.920	2183.62	0.70	923.06	1.07	647.30	0.18	196.49	1.35	148.89	1.14	181.88	0.73
320	80	1.917	2186.45	0.87	926.08	0.51	666.94	0.18	193.16	1.16	148.82	1.12	165.46	0.72
Hard	86	1.912	2192.06	0.61	935.92	1.09	668.03	0.44	201.97	0.91	146.83	0.59	153.27	0.45
Soft	100	1.922	2182.25	0.66	918.56	1.11	622.74	0.92	196.61	1.38	156.98	1.38	199.70	0.87
Panel 2: Removing Initial Lotteries														
Rank	100	1.718	2418.14	0.93	930.67	0.81	1073.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
Receivers	100	2.218	1900.85	1.41	521.00	0.00	1073.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00

optimization framework to fully determine the ordering of students and thus potentially improve priority allocations.

Panel 1 in Table 4 reports the results of applying the Absolute-Hybrid, Hard and Soft approaches under the alternative objective function (see Appendix E.2.1), while Panel 2 shows the results of implementing Absolute-Hard in the absence of initial random tie-breakers when the objective is to minimize the sum of ranks of assignments or to maximize the number of receivers of priority, labeled as *Rank* and *Receivers* for simplicity, respectively. We refer to Appendix E.2.2 for a detailed formulation. Without initial lotteries, a feasible solution for Absolute-Hard is always guaranteed, enabling us to assess the maximum potential benefits of relaxing tie-breaking rules.

Comparing the results in Panel 1 with those in Table 3, we observe that changing the objective function substantially increases the number of siblings assigned together when absolute contingent priorities are soft or when the hybrid approach is implemented with low values of ζ. In contrast, when priorities are strict (Absolute-Hard) or hybrid with high values of ζ, altering the objective has little effect. This finding suggests that the potential gains from optimizing a different objective are limited, consistent with our earlier comparison between SOSM and FOSM. On the other hand, removing initial lotteries has a much larger impact on sibling placement, as shown in Panel 2. In particular, implementing Absolute-Hard without initial lotteries achieves the maximum feasible number of siblings assigned together, regardless of the objective function. However, focusing on the sum of ranks improves the alignment with individual preferences, decreasing the average rank of assignment and increasing the share of students placed in their top choice. These gains, however, come at the cost of more unassigned students, as the number decreases from 930.67 to 521.00 when changing the objective (to maximize receivers of priority). This trade-off is intuitive, as minimizing the sum of ranks of assignment prioritizes matching students to their top choices, leading to a lower average rank and more top-choice assignments, but potentially leaving some students unassigned.

Overall, these results indicate that removing initial lotteries is a powerful policy lever, substantially increasing sibling co-assignments. Yet, the choice of objective function remains crucial: decision makers must weigh access—by optimizing the sum of ranks—against choice—by optimizing the number of receivers of priority—according to their policy goals.

6. Discussion

Our results show that considering absolute contingent priorities can significantly improve the outcomes for students with siblings (e.g., preference of assignment and probability of getting assigned together with their siblings), while having no sizable negative effects on other relevant outcomes such as preference of assignment or number of unassigned students. Moreover, we find that absolute significantly outperforms other benchmarks specially designed to target students with siblings, such as the algorithm currently used in Chile and the stable matching that maximizes siblings assigned together. Most drawbacks of our proposed framework can be mitigated using a hybrid approach that combines the soft version of absolute priorities with a minimum number of effective providers, ensuring the existence of stable matchings while benefiting a large number of students. Therefore, clearinghouses focused on the joint assignment of siblings may largely benefit from implementing this hybrid approach. Indeed, we have collaborated closely with Chilean authorities, who have recognized the value of our hybrid approach and included it in the official list of proposed improvements for the assignment process. However, due to political challenges unrelated to our framework, it has yet to be implemented in practice.

Our work also illustrates the importance of carefully studying different approaches to achieve a specific outcome (e.g., increasing the number of siblings assigned together), as seemingly irrelevant choices may play a substantial role. For instance, our results show that varying the extent to which prioritized students can displace non-prioritized ones leads to entirely different outcomes. Similarly, the choice of tie-breaking rule and whether contingent priorities are hard or soft can have important effects on some properties of the mechanism, such as the existence of a stable matching. Finally, although we focus on school choice as a motivating example, there are many other settings where participants may care about the assignment of others and where clearinghouses may benefit from their joint assignment, including daycare and refugee resettlement. We believe the guidelines and insights derived from our work may help design policies to achieve those outcomes.

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Appendix A: Appendix to Section 3

A.1. Contingent Stability and Other Definitions

In this section, we outline how our definitions of contingent stability compare to other definitions of stability with complementarities. Some of these definitions pertain to a single-level model, where applicants with complementarities apply, with the main application of interest being the market for matching residents to hospitals (Klaus and Klijn 2005, McDermid and Manlove 2010, Biró et al. 2011, Marx and Schlotter 2011, Kojima et al. 2013).[27] Some other definitions of stable matching with complementarities are given when there are siblings applying at different levels of the school system (Correa et al. 2022, Dur et al. 2022, Sun et al. 2023).

Next, we provide four illustrative instances of the matching problem with complementarities to display the key differences between our definitions of contingent stability and those given in the literature. The first three instances are drawn from the survey by Biró and Klijn (2013). For all the instances, we assume that (i) there is only one school c to which all the students apply to, and (ii) all the students are in group $|\mathcal{G}| = 1$.

- Instance 1: There is a family $f = \{f_1, f_2\}$ and a single student s. All the students apply to the same level. School c has two seats and the following preference list: $f_1 \succ s \succ f_2$.
- Instance 2: There are two families, $f = \{f_1, f_2\}$ and $h = \{h_1, h_2\}$. All the students apply to the same level. School c has two seats and the following preference list: $f_1 \succ h_1 \succ h_2 \succ f_2$.
- Instance 3: There are two families, $f = \{f_1, f_2\}$ and $h = \{h_1, h_2\}$. All the students apply to the same level. School c has two seats and the following preference list: $f_1 \succ h_1 \succ f_2 \succ h_2$.
- Instance 4: There is a family $f = \{f_1, f_2\}$ and a single student s, where students s_1 and f_1 apply to level l_1, and student f_2 applies to level l_2. School c has one seat at level l_1 and at level l_2, and the following preference list: $s \succ f_1 \succ f_2$.

In Table 5, we present the matchings that are considered stable according to the different definitions given in the literature and ours. In particular, in the third column, Absolute and Partial stand for contingent stable matching under absolute and partial priorities, respectively. Note also that in instances 1, 2 and 3, the school has only one level, while in Instance 4 the school has two levels. For (single-level) instances 1 to 3, the definitions by McDermid and Manlove (2010) and Sun et al. (2023) coincide, so we use only the first reference as a label in the table.

The first observation is that our definitions of stability, along with the ones employed by Correa et al. (2022) and Sun et al. (2023), are the only ones that identify at least a stable matching for all the instances. Second, the definition given by Dur et al. (2022) can only be used for siblings that are not applying to the same level, while the remaining definition, with exception of Correa et al. (2022), Sun et al. (2023) and ours,

[27] For the work by McDermid and Manlove (2010), we consider the definition of stable matching used for *consistent couples*. Note that Csáji et al. (2023) use the same definition of blocking pair, so the same considerations as in McDermid and Manlove (2010) apply.

Instance	Matching	Stability
1	$\{s\}$	Biró et al. (2011), Kojima et al. (2013)
	$\{f_1, f_2\}$	Absolute, Partial
	$\{s, f_1\}$	Klaus and Klijn (2005), McDermid and Manlove (2010), Marx and Schlotter (2011), Correa et al. (2022)
2	$\{f_1, f_2\}$	McDermid and Manlove (2010), Marx and Schlotter (2011), Kojima et al. (2013), Absolute, Partial
	$\{h_1, h_2\}$	Marx and Schlotter (2011), Kojima et al. (2013), Biró et al. (2011), Absolute
	$\{f_1, h_1\}$	Klaus and Klijn (2005), McDermid and Manlove (2010), Correa et al. (2022), Absolute
3	$\{f_1, f_2\}$	Marx and Schlotter (2011), Kojima et al. (2013), Biró et al. (2011), Absolute, Partial
	$\{h_1, h_2\}$	Kojima et al. (2013), Absolute
	$\{f_1, h_1\}$	Klaus and Klijn (2005), McDermid and Manlove (2010), Correa et al. (2022), Absolute
4	$\{s\}$	Dur et al. (2022)
	$\{f_1, f_2\}$	Correa et al. (2022), Absolute
	$\{s, f_2\}$	Sun et al. (2023), Partial

Table 5 Definitions under which each feasible matching is considered stable

are not given for matching markets with multiple levels. Third, our definitions align with the majority of the existent definitions. In instance 1, this is because our definition allows family f to be matched together, thus encoding as stable a matching in which priorities play a critical role. Instances 2 and 3 display cases in which two families, f and h, are competing for the same positions: In both instances 2 and 3, our definitions of contingent stability consider the matching $\{f_1, f_2\}$ stable; moreover, Absolute identifies as stable all the feasible matchings deemed stable by some other definition from the literature. Therefore, our definition of contingent stable matching under absolute priorities dramatically expands the set of matchings deemed stable in comparison to the other definitions from the literature. Finally, in instance 4, our definition of contingent stable matching under absolute priorities identifies as stable the matching $\{f_1, f_2\}$, in line with the definition by Correa et al. (2022). Sun et al. (2023) treat schools with multiple levels as independent which prevents them to identify as stable the matching $\{f_1, f_2\}$.

A.2. Additional Examples

EXAMPLE 2. Consider an instance with a two families $f = \{f_1, f_2\}$, $f' = \{f_1', f_2'\}$, two levels $\mathcal{L} = \{\ell_1, \ell_2\}$ with $\mathcal{S}^{\ell_1} = \{f_1, f_1'\}$ and $\mathcal{S}^{\ell_2} = \{f_2, f_2'\}$ and a single school c offering one seat in each level. Moreover, suppose that every student prefers c over being unassigned, every student belongs to the same initial group (i.e., $g(s,c) = 1$ for all $s \in \mathcal{S}$), and that the initial vector of random tie-breakers is such that $p_{f_1,c} < p_{f_2',c} < p_{f_1',c} < p_{f_2,c}$. Then, consider the matchings

$$\mu = \left\{(f_1, c), (f_2, c), (f_1', \emptyset), (f_2', \emptyset)\right\}$$

and

$$\mu' = \left\{(f_1, \emptyset), (f_2, \emptyset), (f_1', c), (f_2', c)\right\}.$$

Neither μ nor μ' are initially stable (i.e., using the initial priority order $\succ_c$ induced by the random tie-breakers and omitting contingent priorities). However, if the school considers contingent priorities as an additional group that has higher priority than students with no siblings, then $g^{\mu}(f_2, c) < g^{\mu}(f'_2, c)$ and, thus, $f_2 \succ^{\mu}_c f'_2$. Similarly, $g^{\mu'}(f'_1, c) < g^{\mu'}(f_1, c)$ and, thus, $f'_1 \succ^{\mu'}_c f_1$. As a result, both μ and μ' would be contingent stable given the contingent priority orders $\succ^{\mu}_c$ and $\succ^{\mu'}_c$, respectively. □

EXAMPLE 3. Suppose there are two schools $C = \{c_1, c_2\}$, each offering one seat in each level $\ell \in \mathcal{L} = \{\ell_1, \ell_2\}$. In addition, suppose there are two families $f = \{f_1, f_2\}$, $f' = \left\{f'_1, f'_2\right\}$ and two single child s_1, s_2, where subscripts denote the level to which the student belongs. Finally, suppose that preferences are such that $f_1, f_2, s_2 : c_1 \succ c_2$, $f'_1, f'_2, s_1 : c_2 \succ c_1$, and initial priorities are such that

$$c_1 :\ s_2 \succ_{c_1} f_2 \succ_{c_1} f'_2 \succ_{c_1} f_1 \succ_{c_1} f'_1 \succ_{c_1} s_1$$
$$c_2 :\ f_2 \succ_{c_2} f'_2 \succ_{c_2} s_2 \succ_{c_2} s_1 \succ_{c_2} f'_1 \succ_{c_2} f_1$$

With no contingent priorities, the unique initially stable matching is

$$\mu = \left\{(s_2, c_1), (f_2, c_2), (f_1, c_1), (s_1, c_2), (f'_1, \emptyset), (f'_2, \emptyset)\right\}.$$

In contrast, with contingent priorities, the alternative matching

$$\mu' = \left\{(f_1, c_1), (f_2, c_1), (f'_1, c_2), (f'_2, c_2), (s_1, \emptyset), (s_2, \emptyset)\right\}$$

is stable as long as $g^{\mu'}(f_2, c_1) < g^{\mu'}(s_2, c_1)$ and $g^{\mu'}(f'_1, c_2) < g^{\mu'}(s_1, c_1)$. This condition can hold because f_1 may grant contingent priority to f_2 in c_1, allowing f'_2 to be assigned to c_2 and, consequently, provide contingent priority to f'_1 at that school. However, it is important to note that neither (f'_1, c_2) nor (f'_2, c_2) belong to any stable matching under the initial priority orders $\{\succ_c\}_{c \in C}$.

EXAMPLE 4. Consider an instance with two schools $C = \{c_1, c_2\}$, three levels $\mathcal{L} = \{\ell_1, \ell_2, \ell_3\}$, two single students $\{s, s'\}$ applying to level ℓ_3, two families $f = \{f_1, f_2\}$ and $f' = \left\{f'_1, f'_2, f'_3\right\}$, with students f_1, f'_1 applying to level ℓ_1, f_2, f'_2 applying to ℓ_2, and f'_3 applying to ℓ_3. In addition, suppose that preferences are $f_1, f_2, f'_1, f'_2, f'_3 : c_1 \succ \emptyset$ and $s, s' : c_1 \succ c_2$. Finally, suppose that school c_1 offers one seat in each level, school c_2 offers two seats in level ℓ_3 (and zero in the other levels), and that the clearinghouse uses a single tie-breaking rule at the individual level (i.e., $p_{s,c} = p_s$ for all $c \in C$) with realized values: $p_s < p_{s'} < p_{f_1} < p_{f'_2} < p_{f_2} < p_{f'_1} < p_{f'_3}$. In this case, there are two contingent stable matching under absolute priorities:

$$\mu = \left\{(f_1, c_1), (f_2, c_1), (s, c_1), (s', c_2)(f'_1, \emptyset), (f'_2, \emptyset), , (f'_3, \emptyset)\right\},$$
$$\mu' = \left\{(f_1, \emptyset), (f_2, \emptyset), (s, c_2), (s', c_2)(f'_1, c_1), (f'_2, c_1), (f'_3, c_1)\right\}.$$

These two matchings are weakly optimal for students, i.e., there is at least one student who strictly prefers μ over any other contingent stable matching under absolute priorities; similarly for μ'. Moreover, the set of students matched in each case (and even its cardinality) differs. □

Appendix B: Appendix to Section 3.3

B.1. Existence

As discussed in Roth (2002), stability is a desirable property since it correlates with the long-term success of the matching process. Unfortunately, as we show in Proposition 3 and Theorem 3, a contingent stable matching under absolute priorities may not exist and the decision problem of certifying the (un)existence is NP-complete.

PROPOSITION 3. *A contingent stable matching under absolute priorities may not exist regardless of the tie-breaking rule, even if families are of size at most two and the initial tie-breakers are at the family level.*

Proof of Proposition 3. It is enough to show the result for a single tie-breaking rule at the family level since all the other tie-breakers can be obtained through small perturbations. Consider an instance with four schools, c_1, c_2, c_3, and c_4, and two levels ℓ_1 and ℓ_2. School c_1 has only one seat at level ℓ_2; schools c_2 and c_4 have one seat in each level; school c_3 has only one seat at level ℓ_1. There are four families of students, $f_a = \{a_1, a_2\}$, $f_x = \{x_1\}$, $f_d = \{d_1, d_2\}$, $f_y = \{y_2\}$. Students a_1, x_1, d_1 apply at level ℓ_1, and students a_2, d_2, y_2, apply at level ℓ_2. The preferences of the families (and of each student) are the following, $f_a : c_3 \succ c_4$; $f_x : c_2$; $f_d : c_1 \succ c_2 \succ c_3$; $f_y : c_4 \succ c_1$. Every school has the same tie-breaker which leads to the following initial order $y_2 \succ x_1 \succ d_1 \succ d_2 \succ a_1 \succ a_2$. Finally, $|\mathcal{G}| = 1$.

Note there is only one initial stable matching:

$$\mu = \{(a_1, c_4), (a_2, \emptyset), (x_1, c_2), (d_1, c_3), (d_2, c_1), (y_2, c_4)\}.$$

Notice that the only matchings that may be contingent stable are those that would match a_1, a_2 in school c_4 or d_1, d_2 in school c_2. We study each case separately.

1. Suppose that there is a matching μ' such that $\mu'(c_4) = \{a_1, a_2\}$. In this case, a_2 cannot be an effective priority provider because condition (iii) in Definition 2 is not satisfied since $y_2 \succ_{c_4} a_2$ and y_2 is either unassigned or matched to c_1 in μ', both of which are less preferred than c_4. Therefore, only a_1 could be an effective priority provider (no one applying to ℓ_1 ranks c_4) and a_2 is the receiver. If y_2 is unmatched, that means c_1 is full at level ℓ_2, which happens only if d_2 is matched there, but in this case y_2 would have contingent justified-envy towards d_2 (unless d_2 is an effective priority provider which cannot happen because d_2 does not satisfy condition (iii)). Therefore, y_2 is matched to c_1. Here we have two cases: (i) $\mu'(c_2) = \{x_1, d_2\}$ and $\mu'(c_3) = \{d_1\}$; (ii) $\mu'(c_2) = \{d_1, d_2\}$, $\mu'(c_3) = \emptyset$, x_1 is unmatched. In (i), note that d_2 is an effective priority provider, so d_1 would have contingent justified envy towards x_1. In (ii), note that c_3 is empty which implies μ' is wasteful as a_1 prefers c_3 more than c_4.
2. Assume that there is a matching μ' such that $\mu'(c_2) = \{d_1, d_2\}$. In this case, d_1 cannot be an effective priority provider because x_1 is more preferred than d_1 and it is unassigned, so d_1 does not satisfy condition (iv) in Definition 2. On the other hand d_2 is an effective priority provider since no other

student applying to level ℓ_2 ranks c_2, consequently, d_1 is receiving priority. Since x_1 is unassigned then, from the previous point a_1 and a_2 cannot be together, so a_1 is matched to c_3. Consequently, $\mu'(c_4) = \{y_2\}$ since a_2 cannot compete with y_2. Therefore, c_1 is empty which would be wasteful as d_2 prefers c_1 over c_2. $\square$

PROPOSITION 4. *A contingent stable matching under partial priorities may not exist when the initial tie-breakers are at the individual level, even if families are of size at most two and there at most two levels. In contrast, the existence is guaranteed when tie-breakers are at the family level.*

Proof of Proposition 4. First, we show that a contingent stable matching may not always exist under individual initial tie-breakers. Then, we show that the contingent stability in the partial case coincides with initial stability when we consider family level tie-breakers and, thus, existence is guaranteed.

Individual Level Tie-breakers. There are four schools, c_1, c_2, c_3, and c_4, and two levels ℓ_1 and ℓ_2. At level ℓ_1, schools c_1 and c_3 have one seat, and all the other schools have two seats. At level ℓ_2, c_1 has one seat, and all the other schools have zero seats. There are five families of students, $f_a = \{a_1, a_1'\}$, $f_x = \{x_1\}$, $f_e = \{e_1\}$, $f_d = \{d_1, d_1'\}$, $f_h = \{h_1, h_2\}$. All the students, except for h_2, apply to level ℓ_1. The preferences of the students (which are the same for both levels) are the following, $f_a : c_3 \succ c_4$; $f_x : c_2$; $f_e : c_2$; $f_d : c_1 \succ c_2 \succ c_3$; $f_h : c_4 \succ c_1$. The random tie-breakers are the same for all schools and lead to the following initial order $h_2 \succ d_1 \succ x_1 \succ e_1 \succ d_1' \succ a_1 \succ h_1 \succ a_1'$. Finally, assume $|\mathcal{G}| = 1$.

In this instance, there is only one initial stable matching:

$$\mu = \{(a_1, c_4), (a_2, \emptyset), (x_1, c_2), (e_1, c_2), (d_1, c_1), (d_2, c_3), (h_1, c_4), (h_2, c_1)\}.$$

Notice that the only matchings that may be contingent stable are those that would match a_1, a_1' in school c_4, d_1, d_1' in school c_2, or h_1, h_2 in school c_1. We now study each case separately.

1. Let μ' be a matching such that $\mu'(c_4) = \{a_1, a_1'\}$. It is easy to see that a_1 is the effective priority provider and a_1' is the priority receiver. Due to partial priorities h_1 is either unmatched or matched to c_1. If h_1 is unmatched is because c_1 is full at level ℓ_1 which means d_1 and d_1' are matched there. However, to satisfy contingent stability h_2 must also be matched to c_1 at level ℓ_2 (otherwise it would be wasteful). Clearly, h_2 is an effective priority provider there, so h_1 receives h_2's tie-breaker displacing d_1'. Consequently, d_1' is matched to c_3 (because d_1' cannot compete with x_1 and e_1 in c_2). This leaves one seat free in c_3 which is wasteful because a_1 or a_1' prefer c_3 more than c_4.
2. Let μ' be a matching such that $\mu'(c_2) = \{d_1, d_1'\}$. Note that the only option for this is that h_1 and h_2 are matched to c_1 (if h_1 is matched to c_4 or unmatched is the same reasoning), however, that leaves an empty seat in c_1 which is wasteful as d_1 prefers c_1 over c_2.
3. Let μ' be a matching such that $\mu'(c_1) \supseteq \{h_1, h_2\}$. The only interesting case here is that d_1 is matched to c_1 as well. This case can be analyzed as the first point since we already know that a_1 and a_1' cannot be together.

Family Level Tie-breakers. This result easily follows by noting from Definition 4 that: (i) contingent groups coincide with the initial groups; (ii) $\boldsymbol{p}^{\mu} = \boldsymbol{p}$ for any μ when tie-breakers are at the family level. Therefore, the set of contingent stable matchings under partial priorities coincide with the set of initial stable matchings. □

B.2. Complexity

THEOREM 3. *The problem of determining whether a contingent stable matching with absolute priorities exists is NP-complete, even if each family is of size at most 2, there are at most 2 levels and initial tie-breakers are at the family level.*

Proof. In this section, we use a reduction from the complete stable matching problem with ties and incomplete preference lists (COM-SMTI). This problem consists of two sides (men and women), where each preference list is of size at most 3. Men have strict preferences lists and women have either a strict preference list of size at most 3 or a tie of size 2 as a preference list. The decision problem of whether there exists a complete stable matching in this setting is known to be NP-complete (McDermid and Manlove 2010).

PROPOSITION 5. *COM-SMTI can be reduced in polynomial time to Absolute.*

For this reduction, our main gadget is the example that we use for non-existence in Proposition 3.

REMARK 4. There are two main observations from the example in Proposition 3:

1. Suppose there is an extra school $\hat{c}$ (which offers only level ℓ_2 and has one seat) such that $\hat{c} \succ_{d_2} c_1$. Then, in this new instance, the following is a contingent stable matching under absolute priorities: $\mu(\hat{c}) = \{d_2\}, \mu(c_1) = \{y_2\}, \mu(c_2) = \{x_1\}, \mu(c_3) = d_1, \mu(c_4) = \{a_1, a_2\}$. In words, relative to the initially stable matching given in the proof of Proposition 3, a_2 displaced y_2, who was moved to c_1 thanks to the space that d_2 left (no cycling occurs as in the example).
2. Suppose we add an extra seat to school c_4 in level ℓ_2. In this setting, a_2 can be matched to c_4 which gives a contingent stable matching under absolute priorities.

Proof of Proposition 5. Let I be an instance of COM-SMTI composed by U the set of men, $W = W^s \cup W^t$ the set of women where W^s is the set of women with a strict preference list and W^t the set of women with a tie of length 2 as a preference list. Without loss of generality, we assume that $|U| = |W|$. We denote by $P(u)$ the preference list for $u \in U$, similarly $P(w)$ for $w \in W$. For a women with a tie as a preference list we denote by u the man in the first position and u' the man in the second position (this become relevant in the preference lists in Figure 1).

Instance Construction. We construct an instance I' from a given instance I as follows:

1. For every man $u \in U$, we create 4 dummy schools $c_u^1, c_u^2, c_u^3, c_u^4$, 4 dummy families $f_u^a = \{a_u^1, a_u^2\}, f_u^d = \{d_u^1, d_u^2\}, f_u^x = \{x_u^1\}, f_u^y = \{y_u^2\}$ where the superscripts of the students indicate the level in which the students are applying to. The preference lists and capacity of the schools are in Figure 1. In particular,

the preference list of the schools c_u^j with $j \in [4]$ are all the same. Note that this construction follows the counter-example in Proposition 3.

2. For every women $w \in W^s$ we create a school c_w that only offers level 2 with capacity 1.
3. For every women $w \in W^t$, we create two dummy schools h_w^1, h_w^2 that only offer one seat in level 2, 4 dummy schools $c_w^1, c_w^2, c_w^3, c_w^4$, 5 dummy families $f_w^a = \{a_w^1, a_w^2\}, f_u^d = \{d_w^1, d_w^2\}, f_w^x = \{x_u^1\}, f_w^y = \{y_w^2\}, f_w^{\hat{y}} = \{\hat{y}_w^2\}$ where superscript indicates the level the students are applying to. We present the preference lists and capacity of the schools in Figure 1.

Clearly the construction above can be done in polynomial time. In Figure 1, $P(u)$ corresponds to the original preference list of $u \in U$ with women $w \in W^s$ replaced by c_w and women $w \in W^t$ replaced by either h_w^1 or h_w^2 depending on the position of u in the tie (consider an arbitrary order that applies to all the ties). On the other hand, for each $w \in W$, $P(w)$ corresponds to the original preference list of w with men replaced by d_u^2.

Remark: The main idea of the construction is that the instance of COM-SMTI is embedded in level 2 of schools c_w, h_w^1, h_w^2 (women) and students d_u^2 (men).

$\{a_u^1, a_u^2\}: \ c_u^3 \succ c_u^4$ for all $u \in U$

$x_u^1: \ c_u^2$ for all $u \in U$

$\{d_u^1, d_u^2\}: \ P(u) \succ c_u^1 \succ c_u^2 \succ c_u^3$ for all $u \in U$

$\{y_u^2\}: \ c_u^4 \succ c_u^1$ for all $u \in U$

$c_u^1(0,1), c_u^2(1,1), c_u^3(1,0), c_u^4(1,1): \ y_u^2 \succ x_u^1 \succ d_u^1 \succ d_u^2 \succ a_u^1 \succ a_u^2$ for all $u \in U$

$c_w(0,1): \ P(w)$ for all $w \in W^s$

$\{a_w^1, a_w^2\}: \ c_w^3 \succ c_w^4$ for all $w \in W^t$

$x_w^1: \ c_w^2$ for all $w \in W^t$

$\{d_w^1, d_w^2\}: \ c_w^1 \succ c_w^2 \succ c_w^3$ for all $w \in W^t$

$\{y_u^2\}: \ h_w^1 \succ c_w^4 \succ c_w^1$ for all $w \in W^t$

$\{\hat{y}_u^2\}: \ h_w^2 \succ c_w^4 \succ c_w^1$ for all $w \in W^t$

$h_w^1(0,1): \ d_u^2 \succ y_w^2$ for all $w \in W^t$

$h_w^2(0,1): \ d_{u'}^2 \succ \hat{y}_w^2$ for all $w \in W^t$

$c_w^1(0,1), c_w^2(1,1), c_w^3(1,0), c_w^4(1,2): \ \hat{y}_w^2 \succ y_w^2 \succ x_w^1 \succ d_w^1 \succ d_w^2 \succ a_w^1 \succ a_w^2$ for all $w \in W^t$

Figure 1 Capacities are next to each school in parenthesis where each component indicates the level's capacity.

CLAIM 1. *If M is a complete stable matching in I, then there exists a contingent stable matching in I'.*

Proof. Given a complete matching M in I, we construct a matching μ in I' as follows: For every man $u \in U$ and a women $w \in W$ such that $M(w) = u$, then we match $\mu(c_w) = d_u^2$ if $w \in W^s$ or $\mu(h_w^1) = d_u^2$ if $w \in W^t$ and u is in the first position of tie or, otherwise $\mu(h_w^2) = d_u^2$. The remaining matches are as follows:

$$\mu(c_u^1) = \{y_u^2\}, \quad \mu(c_u^2) = \{x_u^1\}, \quad \mu(c_u^3) = \{d_u^1\}, \quad \mu(c_u^4) = \{a_u^1, a_u^2\},$$

If d_u^2 is matched to h_w^1 then we add the following pairs

$$\mu(h_w^2) = \hat{y}_w^2 \quad \mu(c_w^1) = d_w^2 \quad \mu(c_w^2) = \{x_w^1\} \quad \mu(c_w^3) = \{d_w^1\} \quad \mu(c_w^4) = \{a_w^1, a_w^2, y_w^2\}$$

Otherwise, if d_u^2 is matched to h_w^2 then we add the following pairs

$$\mu(h_w^1) = y_w^2 \quad \mu(c_w^1) = d_w^2 \quad \mu(c_w^2) = \{x_w^1\} \quad \mu(c_w^3) = \{d_w^1\} \quad \mu(c_w^4) = \{a_w^1, a_w^2, \hat{y}_w^2\}$$

Note that since matching M is complete, no two distinct d_u^2, $d_{u'}^2$ can be tentatively matched to the same c_w where $w \in W^s$; moreover, no students in I' are unmatched. Clearly this matching is contingent stable under absolute priorities thanks to the observations in Remark 4. □

CLAIM 2. *Given a contingent stable matching under absolute priorities μ in I', we can construct a complete stable matching M in I.*

Proof. First, we show that in μ a student d_u^2 must be either matched to c_w, h_w^1 or h_w^2 in $P(u)$. Suppose otherwise, then it is matched to c_u^1 but this will contradict the contingent stability of μ in that gadget (in other words, this would lead to the counterexample in Proposition 3).

Now we show that no school created from a woman $w \in W$ will be matched to more than one student of type d_u^2 in μ. First note that for c_w this is trivial as it has capacity 1. Now we need to show that at most one school, between h_w^1 and h_w^2, is matched to a student of the type d_u^2. Suppose the opposite in which h_w^1 is matched to d_u^2 and h_w^2 is matched to $d_{u'}^2$. Then $\hat{y}_w^2$ and y_w^2 would be match to c_w^4. However, a_w^2 can displace y_w^2 by claiming contingent priority thanks to a_w^1. Then y_w^2 would displace d_w^2 (e.g. currently matched to c_w^1) which then will be matched to c_w^2. This will create the same "cycle" that we found in Proposition 3.

In conclusion, every d_u^2 is matched to c_w, h_w^1 or h_w^2. Moreover, schools of type h_w^1 and h_w^2 receive at most one student of the form d_u^2. Matching M is constructed as

$$M = \{(u, w) \in U \times W : \ (d_u^2, c_w) \in \mu \text{ or } (d_u^2, h_w^1) \in \mu \text{ or } (d_u^2, h_w^2) \in \mu\},$$

whose stability result from the contingent stability of μ in those schools which is essentially one level and no siblings apply to those schools (siblings of type d_u^1 only applies to level 1 and those schools do not have that level), therefore, the standard stability. Finally, since every d_u corresponds to a man in I, and every d_u is matched, then M is complete. □

Since Absolute is clearly in NP, we conclude the proof of Proposition 5 and Theorem 3 from both claims. □

THEOREM 4. *The problem of determining whether a contingent stable matching with partial priorities under individual tie-breakers exists is NP-complete, even if the size of each family is at most 2 and there are at most 2 levels.*

Proof. To show this result, it is enough to show that COM-SMTI can be reduced in polynomial time to Partial. The proof follows a similar reasoning as the one for absolute priorities, while using as a pivot model the one from Proposition 4. The main differences are the following:

1. If $u \in U$, then we create dummy schools and students as in Proposition 4, where the preference list of family $f(h)$ is $P(u) \succ c_u^4 \succ c_u^1$.
2. If $w \in W^t$ is such that it has preference list (u, u'), then we create dummy schools and students as in Proposition 4 with the following modifications: (1) we also add two students $y_w^2, \bar{y}_w^2$ and two schools $z_w^2, \bar{z}_w^2$ (each with one spot at level 2); the preference of y_w^2 is $z_w^2 \succ c_w^1$ (similar for $\bar{y}_w^2$); the preference of z_w^2 is $d_u \succ y_w^2$ (similar for $\bar{z}_w^2$). (2) The preference list of the schools $c_w^1, c_w^2, c_w^3, c_w^4$ is $y_w^2 \succ \bar{y}_w^2 \succ h_w^2 \succ d_w^1 \succ x_w^1 \succ e_w^1 \succ d_w^{1'} \succ a_w^1 \succ h_w^1 \succ a_w^{1'}$.

□

B.3. Incentives

PROPOSITION 6. *The mechanism to find a rank-optimal contingent stable matching with absolute priorities is not strategy-proof for the families, regardless of the tie-breaking rule.*

Proof of Proposition 6. First, note that it is enough to show the result for a single tie-breaking rule at the family level since we can construct examples for all the other tie-breaking rules by simply adding a small perturbation to this counter-example.

Consider an instance of the problem with two single students s_1, s_2 and two families $f = \{f_1, f_2\}$, $f' = \{f_1', f_2'\}$, with students s_1, f_1, f_1' applying to level ℓ_1 and s_2, f_2, f_2' applying to level ℓ_2. In addition, suppose there are four schools $C = \{c_1, c_2, c_3, c_4\}$ each offering one seat in each level except for c_2 in ℓ_1 and c_3 in ℓ_2, for which $q_{c_2}^{\ell_1} = q_{c_3}^{\ell_2} = 0$. Suppose that students' preferences are:

$$f_1 : c_1 \succ c_3, \qquad f_2 : c_1 \succ c_2$$
$$f_1' : c_3 \succ c_4, \qquad f_2' : c_3 \succ c_4$$
$$s_1 : c_1 \succ c_2, \qquad s_2 : c_4 \succ c_1.$$

Finally, the clearinghouse uses a single tie-breaker at the family level (i.e., $p_{s,c} = p_{f(s)}$ for all $s \in \mathcal{S}$ and $c \in C$) whose realized values are:

$$p_{s_2} < p_{s_1} < p_f < p_{f'} \quad \Leftrightarrow \quad s_2 \succ s_1 \succ f \succ f'$$

Note that, if every family reports their preferences truthfully, then there is a unique *rank-optimal* contingent stable matching under absolute priorities:

$$\mu = \left\{(s_1,\emptyset),(f_1,c_1),(f_1',c_3),(s_2,c_4),(f_2,c_1),(f_2',\emptyset)\right\}. \tag{9}$$

In this case, four students get assigned to their top choice and two of them get unassigned. Note that f_2' may unilaterally improve their assignment by adding more schools to their reported list. To see this, suppose that f_2' reports the following preference order:

$$f_2' : c_4 \succ c_3 \succ c_1.$$

Based on these new preferences, the matching μ is no longer rank-optimal as it is dominated by the matching

$$\mu' = \left\{(s_1,c_1),(f_1,c_3),(f_1',c_4),(s_2,c_1),(f_2,c_2),(f_2',c_4)\right\}, \tag{10}$$

since four students ($\left\{f_1,f_1',s_2,f_2\right\}$) get assigned to their second choice, two ($\left\{s_1,f_2'\right\}$) get their top choice, and no student results unassigned. Hence, the assignment μ' leads to a strict improvement over the sum of the students' rankings and, thus, f_2' can improve their assignment by misreporting their preferences.

PROPOSITION 7. *The mechanism to find a contingent rank-optimal stable matching with partial priorities is not strategy-proof for the families under individual initial tie-breakers, but it is strategy-proof under family tie-breaking rule.*

Proof of Proposition 7. We first show that the mechanism is not strategy-proof under individual tie-breakers. As before, it is enough to show that this is the case under single tie-breakers, as the result for multiple tie-breakers can be obtained by adding a small perturbation.

Consider the same market as described in the proof of Proposition 6 but with a small variation in the tie-breakers. Specifically, suppose they are given by:

$$p_{f_2} < p_{s_1} < p_{f_1} < p_{f_1'} < p_{s_2} < p_{f_2'} \quad \Leftrightarrow \quad f_2 \succ s_1 \succ f_1 \succ f_1' \succ s_2 \succ f_2'.$$

Then, the assignment in (9) is also the only rank-optimal contingent stable matching under partial priorities. Moreover, as before, f_2' can improve its assignment by misreporting their preferences by including all the schools in the following order:

$$f_2' : c_4 \succ c_1 \succ c_3.$$

In this case, the assignment μ' in (10) that f_2' strictly prefers is also feasible and leads to an overall better assignment if the goal is to find a rank-optimal contingent stable matching under partial priorities.

Finally, the fact that the mechanism returning a rank-optimal contingent stable is strategy-proof under family level tie-breakers is a corollary of Proposition 4 and, specifically, of the equivalence between partial contingent priorities and initial stability.

PROPOSITION 8. *The mechanism to find a contingent stable matching (under either absolute or partial priorities) is strategy-proof in the large.*

Proof of Proposition 8. We prove the result for the mechanism to find a contingent stable matching with absolute priorities with a single tie-breaking rule. The proof for the other cases follows similarly. Azevedo and Budish (2018) show that a sufficient condition for a mechanism to be strategy-proofness in the large is to be (i) *semi-anonymous* and (ii) *envy-freeness but for ties* (EF-TB). Hence, it is enough to show that our mechanism satisfies these two properties.

Semi-anonimity. As defined in Azevedo and Budish (2018), a mechanism is semi-anonymous if there is a partition Θ of the set of students and, within each element of the partition $\theta \in \Theta$, there is a finite set of types T_θ that specifies the set of possible actions for a student with that type. Specifically, if student s belongs to θ and $t \in T_\theta$ is their type, then the set of possible actions that s can take is defined as $A_{\theta,t} \subseteq A_\theta$. In our school choice setting, we can (i) set $\Theta = \{1, \ldots, |\mathcal{G}| + 1\}^C$, i.e., the partition given by the vector of contingent groups that students belong to in each school, (ii) define the types as students' preferences $\succ_s$, and (iii) set the actions to be the list of preferences that students can submit. Then, two students s and s' such that $g(s,c) = g(s',c)$ for all $c \in C$ (i.e., belong to the same group in each school and, thus, to the same partition element $\theta \in \Theta$) and $\succ_s = \succ_{s'}$ (belong to the same type as they share the same preferences) differ only because of their tie-breakers. Note that Θ has finite cardinality (since the number of schools and groups are finite) and that there is a finite set of preference lists $\succ_s$ that a student s can potentially report since the number of schools is finite. Therefore, we know that the number of groups, the number of types, and the set of possible actions for each group and type are finite, so the mechanism is semi-anonymous.

EF-TB. Given a market with n students, a direct mechanism is a function $\Phi^n : T^n \to \Delta(C \cup \{\emptyset\})^n$ that receives a vector of types T (the application list of each student) and returns a (potentially randomized) feasible allocation. In addition, let $u_t(\tilde{c})$ be the utility that a student with type $t \in T_\theta, \theta \in \Theta$ gets from the lottery over assignments $\tilde{c} \in \Delta(C \cup \{\emptyset\})$ (note that, by assumption, two students belonging to the same type have exactly the same preferences and, thus, get the same utility in each school $c \in C \cup \{\emptyset\}$). Then, a semi-anonymous mechanism is envy-free but for tie-breaking if for each n there exists a function $x^n : (T \times [0,1])^n \to \Delta(C \cup \{\emptyset\})^n$ such that

$$\Phi^n(t) = \int_{l \in [0,1]^n} x^n(t,l)\, dl$$

and, for all i, j, n, t and l with $l_i \geq l_j$, and if t_i and t_j belong to the same type, then

$$u_{t_i}\left[x_i^n(t,l)\right] \geq u_{t_i}\left[x_j^n(t,l)\right].$$

In plain words, to show that our mechanism is EF-TB, we need to show that whenever two students that belong to the same type differ in their tie-breakers, then the assignment of the student with the higher lottery cannot be worse than that of the other student. This follows directly from the definition of the contingent priority order $\succ^{\mu}$, since for each group, we know that the clearinghouse breaks ties within each group using the tie-breaking rule. As a result, if two students s and s' belong to the same group, we know that the resulting matching μ satisfies $\mu(s) \succ_s \mu(s')$ if $s \succ_c s'$ for all $c \in C$. Then, for any function x, it is direct that $u_{t_s}\left[x_s^n(t,l)\right] \geq u_{t_s}\left[x_{s'}^n(t,l)\right]$. Hence, we conclude that our mechanism is EF-TB, and therefore it is strategy-proof in the large.

REMARK 5. Note that our notions of absolute/partial and hard/soft only differ on how they update students' contingent groups and lotteries. Thus, once these are defined, the contingent priority order remains fixed and known. For this reason, the proof of strategy-proofness in the large is analogous for any combination of these elements.

Appendix C: Appendix to Section 4

C.1. Technical Lemmas

LEMMA 1. *Given feasible $\mathbf{x}^\star, \mathbf{z}^\star, \mathbf{y}^\star$ for Formulation 21 (resp. Formulation 7), there exists another feasible solution $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$ for Formulation 21 (resp. Formulation 7) where the effective provider of priority of each family is the student satisfying Definition 2 with the most favorable initial tie-breaker at the school. Moreover, $\mathbf{x}^\star = \mathbf{x}^\blacktriangledown$, i.e., the students' ranking has the same value.*

Proof. Let $\mathbf{x}^\star, \mathbf{z}^\star, \mathbf{y}^\star$ be a solution (for either formulation), and suppose that $\exists f \in \mathcal{F}$ with $\{a, a'\} \subseteq f$ where both a, a' satisfy Definition 2, $z^\star_{a,c} = 1$, $z^\star_{a',c} = 0$, and $a' \succ_c a$. Consider the case when $x^\star_{a',c} = 1$ (the converse case is direct). Then, we can always set $y^\star_{a,a'',c} = 1$ for all $a'' \in f(a) \setminus \{a\}$ such that $x^\star_{a'',c} = 1$, as it would not change the objective and would still satisfy all constraints. Then, let $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$ be such that $\mathbf{x}^\blacktriangledown = \mathbf{x}^\star$, $z^\blacktriangledown_{a,c} = 0$, $z^\blacktriangledown_{a',c} = 1$, $z^\blacktriangledown_{s,c'} = z^\star_{s,c'}$ for all $(s, c') \notin \{(a,c),(a',c)\}$, $y^\blacktriangledown_{a,a',c} = 0$, $y^\blacktriangledown_{a',a'',c} = 1$ for all $a'' \in f(a') \setminus \{a'\}$ with $x^\blacktriangledown_{a'',c} = 1$, and $y^\blacktriangledown_{s,s',c'} = y^\star_{s,s',c'}$ for all $s, s' \notin f$, $c' \in C$. Since $\mathbf{x}^\star = \mathbf{x}^\blacktriangledown$, the objective function value remains unchanged. Also, it is easy to see that $\mathbf{x}^\blacktriangledown \in \mathcal{P}$, $\mathbf{z}^\blacktriangledown \in \mathcal{R}(\mathbf{x}^\blacktriangledown)$, and $\mathbf{y}^\blacktriangledown \in \mathcal{R}(\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown)$. Moreover, note that

$$z^\star_{s,c'} + \sum_{s' \in f(s)\setminus\{s\}} y^\star_{s',s,c'} = z^\blacktriangledown_{s,c'} + \sum_{s' \in f(s)\setminus\{s\}} y^\blacktriangledown_{s',s,c'}, \quad \forall s \in \mathcal{S},\ s' \in f(s),\ c' \in C.$$

In the absolute case, this implies that both sets of constraints (6a) and (6b) are directly satisfied by $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$.

In the partial case, note that the right-hand side of (7a) is always at least as high under $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$ compared to $\mathbf{x}^\star, \mathbf{z}^\star, \mathbf{y}^\star$ for pairs (s, c) with $\ell(s) = \ell(a)$ (since $y^\star_{a',a,c} = 0$ and $y^\blacktriangledown_{a',a,c} = 1$), while they are equal if we consider the constraint for (s, c) with $\ell(s) = \ell(a')$ since $1_{\{a' \prec_c s \prec_c a\}} = 0$ as $a' \succ_c a$. Hence, the right-hand side of (7a) is always at least as high under $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$ compared to $\mathbf{x}^\star, \mathbf{z}^\star, \mathbf{y}^\star$. Finally, note that the right-hand side of the set of constraints (7b) is also always at least as high under $\mathbf{x}^\blacktriangledown, \mathbf{z}^\blacktriangledown, \mathbf{y}^\blacktriangledown$ compared to $\mathbf{x}^\star, \mathbf{z}^\star, \mathbf{y}^\star$. To see this, note first that $a \succ_c s \wedge s'$ implies $1_{\{a \prec_c s \wedge s'\}} = 1_{\{a' \prec_c s \wedge s'\}} = 0$, so these constraints are always satisfied. On the other hand, if $a \prec_c s \wedge s'$, we have two cases:

- If s is such that $\ell(s) = \ell(a)$, then $y^{\star}_{a',a,c} \cdot 1_{\{a \wedge a' \succ_c s \wedge s'\}} = 0 \leq y^{\blacktriangledown}_{a',a,c} 1_{\{a \wedge a' \succ_c s \wedge s'\}}$, where the equality holds since $y^{\star}_{a',a,c} = 0$ and $a \prec_c s \wedge s'$, while the inequality follows since $y^{\blacktriangledown}_{a',a,c} = 1$ and potentially $a' \succ_c s \wedge s'$ (which would imply strict inequality).
- If s is such that $\ell(s) = \ell(a')$, then (7b) follows because $a' \succ_c a$ implies that $1_{\{a \wedge a' \succ_c s \wedge s'\}} = 1_{\{a' \succ_c s \wedge s'\}}$ so, if $a' \succ_c s \wedge s'$, then $x_{a',c}$ is multiplied by zero, and if $a' \prec_c s \wedge s'$ would imply that there is no term in the rightmost sum, leading to $x^{\star}_{a,c} = 0$ which is a contradiction to the contingent stability of $\mathbf{x}^{\star}$.

Thus, the right hand side of the set of constraints (7b) is always at least as high under $\mathbf{x}^{\blacktriangledown}, \mathbf{z}^{\blacktriangledown}, \mathbf{y}^{\blacktriangledown}$ compared to $\mathbf{x}^{\star}, \mathbf{z}^{\star}, \mathbf{y}^{\star}$, so these constraints are always satisfied for the former.

LEMMA 2. *Consider a contingent stable matching μ (either under absolute or partial priorities) and a school c. Then, every family of size at least 2 with some member s matched to c and another member $s' \in f(s) \setminus \{s\}$ either matched to c or a less preferred school must have at least one member satisfying the conditions of priority provider given in Definition 2 for school c.*

Proof. Fix a contingent stable matching μ. Suppose by contradiction that there exists a school c and a family f with $|f| \geq 2$ such that $\exists s' \in f(s)$ with $\mu(s') \preceq_{s'} c$ and every sibling $s \in f$ with $\mu(s) = c$ does not satisfy the conditions of priority provider given in Definition 2 for school c. In particular, each $s \in f$ with $\mu(s) = c$ does not satisfy condition (iii) in Definition 2, which means that $\left|\left\{\hat{s} \in \mathcal{S}^{\ell(s)} \,:\, \hat{s} \succ_c s,\; c \succeq_{\hat{s}} \mu(\hat{s})\right\}\right| \geq q_c^{\ell(s)}$. From this, we note that if $\left|\left\{\hat{s} \in \mathcal{S}^{\ell(s)} \,:\, \hat{s} \succ_c s,\; c \succ_{\hat{s}} \mu(\hat{s})\right\}\right| = 0$, then $\left|\left\{\hat{s} \in \mathcal{S}^{\ell(s)} \,:\, \hat{s} \succ_c s,\; c = \mu(\hat{s})\right\}\right| \geq q_c^{\ell(s)}$, which leads to a contradiction due to the capacity of school c in level $\ell(s)$ and the fact that $\mu(s) = c$. Therefore, we must have $\left|\left\{\hat{s} \in \mathcal{S}^{\ell(s)} \,:\, \hat{s} \succ_c s,\; c \succ_{\hat{s}} \mu(\hat{s})\right\}\right| > 0$, in other words, there exists a student $\hat{s}$ such that $\hat{s} \succ_c s,\; c \succ_{\hat{s}} \mu(\hat{s})$. Since there is no effective priority provider in school from family $f(s)$ then $\hat{s} \succ_c^{\mu} s$ (either for absolute or partial priorities) which, along with $c \succ_{\hat{s}} \mu(\hat{s})$, implies that $\hat{s}$ has contingent justified-envy towards s. This contradicts the contingent stability of μ. □

C.2. Absolute Priorities

In this section, we focus on the proof of Theorem 1. Let us denote by $\mathcal{A}$ the set of points $(\mathbf{x}, \mathbf{z}, \mathbf{y})$ that satisfy (21) and by $\mathcal{M}$ the set of contingent stable matchings under absolute priorities. We separate the proof in two parts: In the first part, we show that, for any feasible solution $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathcal{A}$, we can construct a matching $\mu \in \mathcal{M}$ (Appendix C.2.1). In the second part, we show the opposite direction, i.e., for any matching $\mu \in \mathcal{M}$, there exists a feasible point $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathcal{A}$ (Appendix C.2.2). Note that we can safely assume that both sets, $\mathcal{A}$ and $\mathcal{M}$, are not empty (if both are empty the theorem follows and if only one of them is empty, then one of the proof parts below would lead to a contradiction).

C.2.1. Proof of Theorem 1, Part I. Consider a feasible solution $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathcal{A}$. Recall that variables $\mathbf{y}$ and $\mathbf{z}$ are defined only for families of size at least 2 in schools where at least two of their members apply. Construct a matching μ as follows: For every feasible pair $(s, c) \in \mathcal{V}$, set $\mu(s) = c$ if and only if $x_{s,c} = 1$.

Since $\mathbf{x} \in \mathcal{P}$, then μ is a matching. For every student $s \in \mathcal{S}$ and school $c \in \mathcal{C}$ we set values $z^{\mu}(s,c) \in \{0,1\}$ according to the rules given after Definition 2 for matching μ. Now, we show that μ is contingent stable under absolute priorities.

First, let us prove that no student has contingent justified-envy. To find a contradiction, suppose that there exists a student s that has contingent justified-envy towards student a matched to school c (in the same level), i.e., $\mu(a) = c$ (i.e., $x_{a,c} = 1$), $c \succ_s \mu(s)$ (i.e., $x_{s,c'} = 0$ for $c' \succeq_s c$) and $s \succ_c^{\mu} a$. Let us assume that $a \notin f(s)$ as the other case is straightforward. Constraint (6a) for pair (s,c) corresponds to

$$q_c^{\ell(s)} \leq \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)}, \\ a \prec_c s}} y_{a',a,c} + \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \prec_c s}} z_{a,c}. \tag{11}$$

Since $s \succ_c^{\mu} a$, then either (a) $g^{\mu}(s,c) < g^{\mu}(a,c)$ or (b) $g^{\mu}(s,c) = g^{\mu}(a,c)$ and $s \succ_c a$ (i.e., $p_{s,c} < p_{a,c}$).[28] We study each case separately:

(a) Suppose that $g^{\mu}(s,c) < g^{\mu}(a,c)$. Then, we know that s has a sibling s' that is an effective priority provider (i.e., $z^{\mu}(s',c) = 1$) and a either has no siblings assigned to c (i.e., $|f(a) \cap \mu(c)| \leq 1$) or has no effective priority provider ($z^{\mu}(a',c) = 0$ for all $a' \in f(a) \setminus \{a\}$). This implies that $x_{s,c'} = 0$ for $c' \succeq_s c$, $x_{s',c} = 1$, $x_{a,c} = 1$, $z_{a,c} = 0$ and $y_{a',a,c} = 0$ for all $a' \in f(a) \setminus \{a\}$, so constraint (6b) leads to $2 \leq 1$, contradicting the feasibility of $(\mathbf{x},\mathbf{y},\mathbf{z})$.

(b) Assume $g^{\mu}(s,c) = g^{\mu}(a,c)$ and $s \succ_c a$. Then, we have two sub-cases. First, consider the case when both students have no effective priority provider in their families, i.e., $g^{\mu}(s,c) = g^{\mu}(a,c) = |\mathcal{G}| + 1$. Since $s \succ_c a$, then $x_{a,c} = 1$ is not accounted for in any sum in (11), which is a contradiction because it would violate the capacity constraint of school c at level $\ell(s)$. Second, consider the case when both have an effective priority provider (i.e., $g^{\mu}(s,c) = g^{\mu}(a,c) = |\mathcal{G}|$),[29] say s' and a'. Since $x_{s',c} = x_{a,c} = 1$ and $x_{s,c'} = 0$ for $c' \succeq_s c$ then constraint (6b) would result in $2 \leq 1$ because $s \succ_s a$.

Now, we show that μ is non-wasteful. Suppose the contrary, namely there exists a student s and a school c such that $c \succ_s \mu(s)$ and $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$. From $c \succ_s \mu(s)$, we can conclude that constraint (6a) for pair (s,c) results in constraint (11) which contradicts $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$. This finishes the first part of the proof of Theorem 1. □

C.2.2. Proof of Theorem 1, Part II. Consider a contingent stable matching under absolute priorities $\mu \in \mathcal{M}$. We now construct a point $(\mathbf{x},\mathbf{y},\mathbf{z})$ and show that it is in $\mathcal{A}$. For $\mathbf{x}$, we set values $x_{s,c} = 1$ if and only if $\mu(s) = c$ for every $(s,c) \in \mathcal{V}$. For every effective priority provider s in school c with at least one of their siblings assigned to c (i.e., $z^{\mu}(s,c) = 1$ and $|f(s) \cap \mu(c)| \geq 2$), we set $z_{s,c} = 1$. Finally, for every sibling $s' \in f(s) \setminus \{s\}$ such that $\mu(s') = c$, we set $y_{s,s',c} = 1$. Clearly $\mathbf{x} \in \mathcal{P}$ because μ is a matching, $\mathbf{z} \in \mathcal{R}(\mathbf{x})$

[28] Recall that for absolute priorities $\boldsymbol{p}^{\mu} = \boldsymbol{p}$.

[29] Recall Remark 2.

because $z_{s,c} = 1$ only for effective priority providers (satisfying the conditions in Definition 2) with at least one other sibling assigned to c, and one can easily check that $\mathbf{y} \in Q(\mathbf{x}, \mathbf{z})$. We now prove that $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ satisfies constraints (6a) and (6b).

First, suppose that there exists a pair $(s, c) \in \mathcal{V}$ such that constraint (6a) is not satisfied, i.e.,

$$q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) > \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)}, \\ a \prec_c s}} y_{a',a,c} + \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \prec_c s}} z_{a,c}. \tag{12}$$

Since variables are non-negative, (12) implies that $x_{s,c'} = 0$ for $c' \succeq_s c$ (i.e., $c \succ_s \mu(s)$) and the right hand side sums strictly less than $q_c^{\ell(s)}$. There are two cases: First, if $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$, then matching μ would be wasteful (because $c \succ_s \mu(s)$) which contradicts its contingent stability. Second, if $|\mu^{\ell(s)}(c)| = q_c^{\ell(s)}$, (12) implies that the right-hand side does not count all the students in school c. In other words, there is a student a such that $x_{a,c} = 1$ and $a \prec_c s$ who is not receiving priority and is not an effective priority provider with another sibling assigned to c. That means $g^\mu(s, c) = g^\mu(a, c)$ and $p^\mu_{a,c} = p_{a,c} > p_{s,c} = p^\mu(s, c)$, in other words s has justified envy towards a which is a contradiction to the contingent stability of μ.

We now focus on showing that constraints (6b) are satisfied. Suppose there exist a school $c \in C$, family $|f| \geq 2$, siblings $\{s, s'\} \subseteq f$ and student $a \in \mathcal{S}^{\ell(s)} \setminus f$ such that constraint (6b) is not satisfied, i.e.,

$$x_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) > 2 - x_{a,c} + 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c}\right). \tag{13}$$

We have several cases:

(a) If $x_{s',c} \in \{0, 1\}$ and $x_{s,c'} = 1$ for some $c' \succeq_s c$, then (13) would result in

$$x_{s',c} > 2 - x_{a,c} + 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c}\right),$$

which is not feasible under any variable assignment.

(b) If $x_{s',c} = 0$ and $x_{s,c'} \in \{0, 1\}$, we achieve the same conclusion as in (a).

(c) If $x_{s',c} = 1$ and $x_{s,c'} = 0$ for all $c' \succeq_s c$, Lemma 2 and $x_{s',c} = 1$ imply that there is an effective priority provider in family $f(s) = f(s')$ matched to c. We obtain the following

$$x_{a,c} > 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c}\right).$$

Since $x_{a,c} \in \{0, 1\}$, the right-hand side must be equal to 0 to ensure that the constraint is feasible, so we must have that either $a \prec_c s$ or the term in parenthesis is equal to zero:

- If $a \prec_c s$, then we know that $g^\mu(s, c) \leq g^\mu(a, c)$ (since s has an effective priority provider) and $p(s, c) < p(a, c)$ (since $a \prec_c s$), so we know that $s \succ_c^\mu a$. Hence, s has justified-envy towards a, which contradicts the contingent stability of μ.

- If $z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} = 0$, then we know that $g^\mu(s,c) < g^\mu(a,c)$, which in turn implies that $s \succ_c^\mu a$, resulting in the same conclusion as before.

This finishes the second, and final, part of the proof of Theorem 1. □

C.3. Partial Priorities

As we did for absolute priorities in Appendix C.2, we divide the analysis in two parts. Let us denote by $\mathcal{A}$ the set of points $(\mathbf{x}, \mathbf{z}, \mathbf{y})$ that satisfy (7) and by $\mathcal{M}$ the set of contingent stable matchings under partial priorities.

C.3.1. Proof of Theorem 2, Part I. The proof is along the same lines than the proof for absolute priorities, but for completeness we include every argument. Consider a feasible solution $(\mathbf{x}, \mathbf{y}, \mathbf{z}) \in \mathcal{A}$. Construct a matching μ as follows: For every feasible pair $(s,c) \in \mathcal{V}$, set $\mu(s) = c$ if and only if $x_{s,c} = 1$. Since $\mathbf{x} \in \mathcal{P}$, then μ is a matching. For every student $s \in \mathcal{S}$ and school $c \in \mathcal{C}$ we set values $z^\mu(s,c) \in \{0,1\}$ according to the rules given after Definition 2 for matching μ. Now we show that μ is contingent stable under partial priorities.

First, let us prove that no student has contingent justified-envy. To find a contradiction, suppose that there exists a student s that has contingent justified-envy towards student a matched to school c (in the same level) which means that: $\mu(a) = c$ (i.e., $x_{a,c} = 1$), $c \succ_s \mu(s)$ (i.e., $x_{s,c'} = 0$ for $c' \succeq_s c$) and $s \succ_c^\mu a$. Constraint (7a) for pair (s,c) corresponds to

$$q_c^{\ell(s)} \leq \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)} \\ a \prec_c s \prec_c a'}} y_{a',a,c}. \tag{14}$$

Since $s \succ_c^\mu a$, then $g^\mu(s,c) = g^\mu(a,c)$ and $p_{s,c}^\mu < p_{a,c}^\mu$.[30] First, note that s cannot be an effective priority provider in c because $x_{s,c} = 0$. Therefore, we have the following cases:

- Student a is an effective priority provider. Then, $p_{s,c}^\mu < p_{a,c}^\mu = p_{a,c} - \epsilon$ which means either: $s \succ_c a$ or s has an effective priority provider in c, say s', such that $s \wedge s' \succ_c a$. First, if $s \succ_c a$ then we would have a capacity contradiction in school c because the term $x_{a,c} = 1$ is not accounted for in (14). If $s' \succ_c a \succ_c s$, then let us analyze constraint (7b). Since $x_{s,c'} = 0$ for $c' \succeq_s c$, $x_{s',c} = 1$, $x_{a,c} = 1$ and $s' \succ_c a \succ_c s$ we obtain $2 \leq 2 - x_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}}$, which is not satisfied since $x_{a,c} = 1$ and a is not a priority receiver so the right-hand side is 1.

For the next cases, we assume that a is not an effective priority provider:

- Both s and a are not receiving priority in c. In this case, $p_{s,c}^\mu < p_{a,c}^\mu$ is equivalent to $p_{s,c} < p_{a,c}$, i.e., $s \succ_c a$ according to initial orders. This means that $x_{a,c} = 1$ is not accounted in any sum term in (14) as a is not receiving or providing priority, which contradicts the capacity of school c in level $\ell(s)$.

[30] Recall that under partial priorities a student receiving contingent priority obtains the most favorable random tie-breaker between the provider and the receiver.

• Student s is receiving priority (assume from s') and a is not receiving. In this case, $p^{\mu}_{s,c} < p^{\mu}_{a,c} = p_{a,c}$ which leads to $s \wedge s' \succ_c a$. We can conclude similarly to the first point.

• Both s and a are receiving priority, say from s' and a' respectively. Then, $p^{\mu}_{s,c} < p^{\mu}_{a,c}$ means that $s \wedge s' \succ_c a \wedge a'$. Note that this implies $s \wedge s' \succ_c a$ which means that constraint (7b) results in $2 \leq 2 - x_{a,c} + 0$, which is not feasible because $x_{a,c} = 1$.

• Student s is not receiving priority and a is receiving from a'. Then, $p^{\mu}_{s,c} < p^{\mu}_{a,c}$ means that $s \succ_c a \wedge a'$ which leads to the same contradiction as above.

From the above, we conclude that no students in μ have justified-envy.

Now, we show that μ is non-wasteful. Suppose the contrary, namely there exists a student s and a school c such that $c \succ_s \mu(s)$ and $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$. From $c \succ_s \mu(s)$, we can conclude that constraint (7a) for pair (s,c) results in constraint (14) which contradicts $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$. This finishes the first part of the proof of Theorem 2. □

C.3.2. Proof of Theorem 2, Part II. We proceed in the same way as we did for absolute priorities. Consider a contingent stable matching under partial priorities $\mu \in \mathcal{M}$. We now construct a point $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ and show that it is in $\mathcal{A}$. For $\mathbf{x}$, we set values $x_{s,c} = 1$ if and only if $\mu(s) = c$ for every $(s,c) \in \mathcal{V}$. For every effective priority provider s in school c (i.e., $z^{\mu}(s,c) = 1$) with at least one sibling assigned to c (i.e., $|f(s) \cap \mu(c)| \geq 2$), we set $z_{s,c} = 1$. Finally, for every sibling $s' \in f(s) \setminus \{s\}$ such that $\mu(s') = c$, we set $y_{s,s',c} = 1$. Clearly $\mathbf{x} \in \mathcal{P}$ because μ is a matching, $\mathbf{z} \in \mathcal{R}(\mathbf{x})$ because $z_{s,c} = 1$ only for effective priority providers (they satisfy the conditions in Definition 2), and one can easily check that $\mathbf{y} \in \mathcal{Q}(\mathbf{x}, \mathbf{z})$. We now prove that $(\mathbf{x}, \mathbf{y}, \mathbf{z})$ satisfies constraints (7a) and (7b).

First, suppose that there exists a pair $(s,c) \in \mathcal{V}$ such that constraint (7a) is not satisfied, i.e.,

$$q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) > \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)} \\ a \prec_c s \prec_c a'}} y_{a',a,c}. \tag{15}$$

Since variables are non-negative, (15) implies that $x_{s,c'} = 0$ for $c' \succeq_s c$ (i.e., $c \succ_s \mu(s)$) and the right hand side sums strictly less than $q_c^{\ell(s)}$. There are two cases: First, if $|\mu^{\ell(s)}(c)| < q_c^{\ell(s)}$, then matching μ would be wasteful (because $c \succ_s \mu(s)$) which contradicts its contingent stability. Second, if $|\mu^{\ell(s)}(c)| = q_c^{\ell(s)}$ then (15) implies that the right-hand side does not count all the students in school c. In other words, there is a student a such that $x_{a,c} = 1$ and $s \succ_c a$ (i.e., $p_{a,c} > p_{s,c}$) who is not receiving priority. This means that $g^{\mu}(s,c) = g^{\mu}(a,c)$ and $p^{\mu}_{a,c} = p_{a,c} > p_{s,c} = p^{\mu}_{s,c}$. Therefore, we conclude that s has justified envy towards a (recall that $c \succ_s \mu(s)$) which is a contradiction to the contingent stability of μ.

We now focus on showing that constraints (7b) are satisfied. Suppose there exist a school $c \in C$, family $|f| \geq 2$, siblings $\{s, s'\} \subseteq f$ and student $a \in \mathcal{S}^{\ell(s)} \setminus f$ such that constraint (7b) is not satisfied, i.e.,

$$x_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) > 2 - x_{a,c} \cdot 1_{\{a \prec_c s \wedge s'\}} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}}. \tag{16}$$

We have several cases:

(a) If $x_{s',c} \in \{0,1\}$ and $x_{s,c'} = 1$ for some $c' \succeq_s c$, then (15) would result in

$$x_{s',c} > 2 - x_{a,c} \cdot 1_{\{a \prec_c s \wedge s'\}} + \sum_{a' \in f(a)\setminus\{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}},$$

(b) If $x_{s',c} = 0$ and $x_{s,c'} \in \{0,1\}$, we obtain the same conclusion as in (a).

(c) If $x_{s',c} = 1$ and $x_{s,c'} = 0$ for all $c' \succeq_s c$, we know from Lemma 2 that s has at least one effective priority provider. Suppose that s' is s's effective priority provider. Therefore, (16) results in

$$x_{a,c} \cdot 1_{\{a \prec_c s \wedge s'\}} > \sum_{a' \in f(a)\setminus\{a\}} y_{a',a,c} \cdot 1_{\{a' \wedge a \succ_c s' \wedge s\}}.$$

Since $x_{a,c} \in \{0,1\}$, this constraint is valid if and only if $x_{a,c} = 1$, $a \prec_c s' \wedge s$ and the right-hand side is 0, i.e., a is not a priority receiver or $1_{\{a' \wedge a \succ_c s' \wedge s\}} = 0$. Either one implies that $p^{\mu}_{s,c} < p^{\mu}_{a,c} = p_{a,c}$, i.e., $s \succ^{\mu}_c a$, which leads to a contradiction because s would have contingent justified-envy towards a which is not possible due to the contingent stability of μ.

This finishes the second, and final, part of the proof of Theorem 2. □

C.3.3. Hard vs. Soft Absolute Priorities.

Proof of Proposition 1. Consider Formulation (21) with (6b) replaced by (7d). If we set $\mathbf{z} = \mathbf{0}$, then $\mathbf{y} = \mathbf{0}$ and constraints (7d) are always satisfied, no matter $\mathbf{x}$. Then, constraints (6a) reduce to constraints (3a) which captures initial stability. On the other hand, to capture contingent stable matchings with hard absolute priorities we set values $\mathbf{z}$ as we did in Appendix C.2.2. □

Appendix D: Appendix to Section 5

D.1. Benchmarks

D.1.1. Descending. As discussed in Section 5.2.2, the algorithm currently used in Chile processes sequentially in decreasing order (starting from 12th level), updating sibling priorities before moving to the next levels to account for the allocation of older siblings. As shown in Correa et al. (2022), *Descending* obtains a stable assignment if the preferences of families satisfy *higher-first*, i.e., each family prioritizes the assignment of their oldest member. However, this is not the case if some families' preferences do not satisfy this condition. Furthermore, this algorithm may result in a matching that is not stable under contingent priorities and may even fail to find a matching that is rank-optimal, as the following example illustrates.

EXAMPLE 5. Consider an instance with two levels, $\ell_1 < \ell_2$, one school in level ℓ_1 with capacity $q_1^{\ell_1} = 2$, n schools in level ℓ_2 with capacities $q_j^{\ell_2} = 1$ for $j \in \{1,\dots,n-1\}$ and $q_n^{\ell_2} = n+2$, two families $f = \{f_1, f_2\}$ and $f' = \{f_1', f_2'\}$ with students $\{f_i, f_i'\}$ applying to level ℓ_i for $i \in \{1,2\}$, and $n-1$ additional single students (i.e., with no siblings), $s_1,\dots,s_{n-1}$, applying to level ℓ_2. Finally, we assume that the clearinghouse uses a single tie-breaking rule at the individual level (i.e., $p_{s,c} = p_s$ for all s and c) with $p_{f_1} < p_{f_1'} < p_{s_1} < \ldots < p_{s_{n-1}} < p_{f_2} < p_{f_2'}$, and students' preferences are:

$$f_1 : c_1 \succ c_2 \succ \ldots \succ c_n \qquad f_1' : c_2 \succ c_1 > \ldots \succ c_n$$
$$f_2 : c_1 \succ c_2 \succ \ldots \succ c_n \qquad f_2' : c_2 \succ c_1 \succ \ldots \succ c_n$$
$$s_i : c_i \succ c_{i+1} \succ c_{i+2}, \quad \forall i \in \{1, \ldots, n-2\} \quad s_{n-1} : c_{n-1} \succ c_n \quad s_n : c_n$$

Under Descending, level ℓ_2 is processed first. Since no student has older siblings assigned, the resulting matching is

$$\mu = \left\{(f_1, c_1), (f_1', c_2), (f_2, c_n), (f_2', c_n), (s_1, c_1), (s_2, c_2), \ldots, (s_n, c_n)\right\}.$$

In contrast, it is easy to see that the unique stable matching under contingent priorities (both absolute and partial), μ', is equivalent to the one that would be obtained under the *Ascending* algorithm (i.e., processing levels sequentially in increasing order), i.e.,

$$\mu' = \left\{(f_1, c_1), (f_1', c_2), (f_2, c_1), (f_2', c_2), (s_1, c_3), (s_2, c_4), \ldots, (s_n, c_n)\right\}.$$

In the former case, $n-2$ students ($\{s_i\}_{i=1}^n$) are assigned to their top preference and two students (f_2, f_2') are assigned to their nth preference, so the overall sum of preference of assignments in level ℓ_2 is $(n-2) + 2n = 3n-2$. In the latter case, $n-2$ students ($\{s_i\}_{i=1}^n$) are assigned to their third preference, and two students (f_2, f_2') are assigned to their top preference, so the overall sum of preference of assignments is $3(n-2) + 2 = 3n-4$. Since the assignment in level ℓ_1 is the same in both cases, we conclude that μ is not rank-optimal. Furthermore, note that μ is not stable under any notion of contingent priorities (either absolute or partial). □

D.1.2. Descending: Math Programming Formulation As part of the argument of why our framework is preferable to the current practice (e.g., Descending), we can show that the latter can also be implemented as a mathematical program with a similar structure to the one we propose. The main difference is that Descending (i) does not allow students to receive contingent priority from siblings in lower levels, and (ii) does not allow to prioritize providers over students with a higher initial order. The former condition follows directly from the rules of the system, while the latter is a consequence of the fact that Descending processes each level in isolation considering the updates priorities resulting from processing the upper levels but without taking into account the resulting changes in priorities in lower ones.

Specifically, the outcome of the Descending mechanism can be computed by solving the following mathematical program:

$$\min \quad \sum_{(s,c)\in\mathcal{V}} r_{s,c} \cdot x_{s,c}$$

$$s.t. \quad q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c'\in C: \\ c'\succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{a\in\mathcal{S}^{\ell(s)}: \\ a\succ_c s}} x_{a,c} + \sum_{\substack{f\in\mathcal{F}: \\ |f|\geq 2}} \sum_{\substack{\{a,a'\}\subseteq f: \\ a\in\mathcal{S}^{\ell(s)}, \\ a\prec_c s}} y_{a',a,c}\,, \quad \forall (s,c)\in\mathcal{V}, \tag{17a}$$

$$x_{s',c} \cdot 1_{\{\ell(s')>\ell(s)\}} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq 2 - x_{a,c} + 1_{\{a \succ_c s\}} \cdot \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c},$$
$$\forall c \in C, f \in \mathcal{F}, \{s, s'\} \subseteq f, a \in \mathcal{S}^{\ell(s)} \setminus f, \tag{17b}$$
$$\mathbf{x} \in \mathcal{P},\ \mathbf{z} \in \mathcal{R}(\mathbf{x}),\ \mathbf{y} \in Q'(\mathbf{x}, \mathbf{z}). \tag{17c}$$

where $Q'(\mathbf{x}, \mathbf{z})$ restricts $Q(\mathbf{x}, \mathbf{z})$ by adding the set of constraints $y_{s,s',c} \leq 1_{\{\ell(s)>\ell(s')\}}$, $\forall s, s', c$. As previously mentioned, there are two key differences relative to (21). On the one hand, the first term on the left-hand side of (17b) is multiplied by $1_{\{\ell(s')>\ell(s)\}}$, which ensures that the constraint is only active when student s has a sibling in a higher level that is also assigned to school c (capturing point (i) mentioned above). On the other hand, the terms involving $z_{a,c}$ in (6a) and (6b) are not present in (17a) and (17b), respectively, enabling to capture point (ii) mentioned above.

D.1.3. Family Optimal Stable Matching (FOSM). A natural benchmark for comparing our approaches is to find the stable matching that maximizes the number of families whose members are assigned to the same school. The following mathematical programming formulation aims to model this baseline:

$$\max_{\mathbf{t} \in \{0,1\}^{\mathcal{F} \times C}, \mathbf{x} \in \mathcal{P}} \quad \sum_{f \in \mathcal{F}} \sum_{c \in C} \left(\sum_{s \in f} x_{s,c} - |f| \cdot t_{f,c} \right) \tag{18a}$$
$$st. \quad \text{Constraint (3a)}$$
$$\frac{\sum_{s \in f} x_{s,c}}{|f|} \leq t_{f,c} \leq \sum_{s \in f} x_{s,c}\ , \ \forall f \in \mathcal{F}, \forall c \in C. \tag{18b}$$

This formulation is similar to Formulation (3). However, in Formulation (18), we have a new binary variable $t_{f,c}$ which is 1 if and only if family f has at least one sibling in school c, and zero otherwise; this is enforced through constraint (18b). In addition, in the objective, we maximize the number of family members in the same school.

D.2. Descriptive Statistics

Table 6 Summary Statistics by Region, Admissions Process 2023-2024

	Schools		Students		Siblings	
Region	N	Seats	N	Applications	N	%
Arica	919	13169	7566	24401	2015	0.266
Tarapaca	1345	18653	16577	56613	5962	0.360
Antofagasta	1477	28428	24335	86855	6978	0.287
Atacama	1189	20603	11910	37910	3252	0.273
Coquimbo	4800	54033	27892	101412	8605	0.309
Valparaíso	8476	111273	52747	170340	15113	0.287
OHiggins	4908	67203	30752	93230	7736	0.252
Maule	6281	80627	35607	116467	8845	0.248
Nuble	3349	43759	14461	41824	3248	0.225
Biobio	7466	110395	46596	156956	12302	0.264
Araucania	8105	91032	31930	91032	7396	0.232
Rios	3240	38175	12505	34465	3032	0.242
Lagos	6590	73535	27054	85040	6917	0.256
Aysen	721	9330	3293	8886	782	0.237
Magallanes	671	9540	5113	16035	1382	0.270
Metropolitana	20063	357793	200539	726155	57185	0.285
Overall	79600	1127548	536353	1847621	149653	0.279

Note. The values in the last column (Siblings [%]) indicate the proportion of students who have a sibling concurrently applying to the system, i.e., $\text{Siblings}[N]/\text{Students}[N]$.

D.3. Results by Tie-Breaking Rule

Table 7 Results: Multiple Tie-breaking at Individual Level (MTB)

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	74	1.904	2205.66	1.45	940.32	1.14	685.04	1.23	193.93	0.79	139.20	0.89	153.61	0.80
Absolute - Soft	100	1.892	2203.93	1.41	940.84	0.92	550.13	1.14	193.27	0.75	206.59	0.76	229.84	1.41
Partial - Hard	100	1.909	2192.37	1.34	942.66	0.72	545.31	0.93	192.81	0.86	201.76	1.06	241.26	1.16
Partial - Soft	100	1.902	2193.80	1.28	942.34	0.85	478.17	0.95	192.29	0.76	229.40	1.31	284.64	1.32
FOSM	100	1.910	2182.74	1.22	942.18	0.90	418.72	0.76	192.36	0.93	246.96	1.23	332.21	1.39
SOSM	100	1.910	2182.76	1.22	942.18	0.90	418.72	0.76	192.36	0.93	246.96	1.23	332.21	1.39
Descending	100	1.910	2191.24	1.11	942.66	0.84	518.41	1.06	194.81	0.92	209.55	1.00	258.88	1.34
Ascending	100	1.907	2190.66	1.38	942.11	0.96	580.58	0.89	194.56	0.51	184.84	0.62	219.03	0.95

Table 8 Results: Multiple Tie-breaking at Family Level (MTB-F)

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	86	1.910	2193.77	0.81	937.94	1.04	664.43	0.33	203.66	0.83	147.52	0.66	153.78	0.56
Absolute - Soft	100	1.899	2196.49	1.00	941.73	1.02	569.39	0.95	200.41	1.29	201.06	1.55	208.32	1.06
Partial - Hard	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
Partial - Soft	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
FOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
SOSM	100	1.910	2188.19	0.98	942.76	1.04	499.92	1.94	200.71	1.29	225.08	1.53	263.96	1.44
Descending	100	1.910	2192.33	0.95	943.80	1.03	556.84	1.78	200.95	1.30	202.88	1.17	224.69	1.29
Ascending	100	1.911	2189.68	0.62	940.24	0.90	609.85	0.67	199.11	1.25	178.64	1.44	186.76	1.09

Table 9 Results: Single Tie-breaking at Individual Level (STB)

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	99	1.865	2296.55	1.20	955.95	0.80	684.39	1.07	189.24	1.00	147.60	1.06	146.62	1.06
Absolute - Soft	100	1.846	2313.49	0.92	961.35	0.81	543.46	1.42	187.17	0.90	217.90	1.53	227.88	1.44
Partial - Hard	100	1.860	2308.98	0.93	963.74	0.75	551.92	1.36	188.17	0.96	207.89	1.10	228.88	1.05
Partial - Soft	100	1.852	2311.99	0.94	963.67	0.77	479.13	1.27	185.53	0.78	241.72	1.25	274.86	1.53
FOSM	100	1.860	2303.84	0.66	962.66	0.78	420.15	1.18	181.09	0.80	268.22	1.83	315.92	1.69
SOSM	100	1.860	2303.84	0.66	962.66	0.78	420.15	1.18	181.09	0.80	268.22	1.83	315.92	1.69
Descending	100	1.864	2301.39	0.87	960.47	0.51	526.74	1.02	186.75	0.91	218.51	1.70	247.31	1.54
Ascending	100	1.864	2293.89	0.83	960.40	0.88	581.56	1.38	183.77	0.91	200.79	1.48	208.88	1.45

Table 10 Results: Single Tie-breaking at Family Level (STB-F)

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	91	1.860	2304.47	1.22	960.46	1.08	657.16	1.57	208.09	1.15	150.98	1.16	152.20	1.66
Absolute - Soft	100	1.847	2314.35	1.17	964.21	1.26	560.53	1.63	207.50	0.89	197.67	1.38	208.81	1.54
Partial - Hard	100	1.856	2306.84	1.22	966.48	1.15	483.55	1.58	209.64	0.85	220.60	1.04	271.62	1.80
Partial - Soft	100	1.856	2306.84	1.22	966.48	1.15	483.55	1.58	209.64	0.85	220.60	1.04	271.62	1.80
FOSM	100	1.856	2306.84	1.22	966.48	1.15	483.55	1.58	209.64	0.85	220.60	1.04	271.62	1.80
SOSM	100	1.856	2306.84	1.22	966.48	1.15	483.55	1.58	209.64	0.85	220.60	1.04	271.62	1.80
Descending	100	1.860	2308.85	1.11	962.53	1.10	544.95	1.40	209.12	0.91	193.49	1.14	230.80	1.54
Ascending	100	1.860	2301.25	1.30	963.76	1.23	604.63	1.48	209.11	0.97	174.36	1.08	184.69	1.74

D.4. Results by Region

Table 11 Results: Atacama Region

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	85	1.817	5615.34	2.39	1708.15	1.69	1502.22	1.46	265.28	0.91	251.31	1.59	418.28	2.14
Absolute - Soft	100	1.801	5641.78	4.95	1706.66	1.59	1254.82	2.42	262.69	0.75	382.20	1.67	550.47	1.95
FOSM	100	1.820	5575.30	2.43	1704.90	1.32	1022.10	2.05	264.40	0.65	440.10	2.82	745.60	2.12
SOSM	100	1.820	5575.90	2.38	1704.90	1.32	1022.70	2.03	264.40	0.65	440.10	2.82	745.00	2.06
Descending	100	1.821	5577.70	2.52	1703.10	1.35	1206.70	1.97	262.40	0.77	361.30	1.38	611.70	1.22
580	100	1.802	5638.73	2.6	1703.25	1.44	1292.12	2.38	262.57	0.82	367.13	1.39	533.05	2.12
590	100	1.802	5638.61	2.49	1703.33	1.47	1296.92	2.07	263.13	0.77	364.19	1.35	530.71	2.08
600	100	1.802	5639.09	2.49	1702.37	1.5	1301.55	2.16	262.87	0.8	362.15	1.53	528.06	1.83
610	100	1.802	5640.13	2.51	1702.94	1.47	1308.05	1.74	262.97	0.8	359.68	1.39	524.37	1.57
620	100	1.802	5639.57	2.47	1703.15	1.48	1317.8	1.35	263.4	0.8	354.22	1.4	519.5	1.73

Table 12 Results: O'Higgins Region

									Separated					
			Top Pref.		Unassigned		Together		None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	70	1.654	16339.76	5.31	3322.64	3.08	3878.51	4.73	426.71	2.25	493.50	3.51	900.40	2.09
Absolute - Soft	100	1.643	16401.98	4.86	3336.33	2.74	3311.89	3.77	428.11	1.74	761.95	3.76	1235.97	2.79
FOSM	100	1.658	16262.61	4.37	3328.00	2.98	2726.50	3.66	426.20	1.51	907.10	3.94	1716.10	1.75
SOSM	100	1.658	16264.30	4.40	3328.00	2.98	2726.90	3.68	426.20	1.51	906.70	3.89	1716.10	1.75
Descending	100	1.658	16269.60	3.90	3316.10	2.89	3090.80	2.78	415.10	1.61	761.40	2.46	1474.40	1.74
1500	100	1.644	16399.32	4.91	3334.95	2.68	3359.17	3.77	424.98	1.88	752.4	3.16	1204.83	2.24
1600	100	1.645	16395.7	5.15	3326.46	2.5	3427.17	2.75	422.79	1.86	714.13	2.43	1174.69	1.75
1700	100	1.645	16400.34	5.11	3317.55	2.45	3557.62	1.99	420.74	1.77	656.5	2.43	1100.86	2.19
1800	100	1.648	16378.88	5.22	3314.64	2.44	3744.68	0.69	418.53	2.11	579.79	2.35	1001.76	2.94
1900	40	1.65	16367.25	6.98	3319.55	1.75	3938.62	0.58	423.75	3.56	486.62	1.88	900.82	2.54

Table 13 Results: Los Lagos Region

			Top Pref.		Unassigned		Together		Separated					
									None		One		Both	
	Solved	Avg. Pref.	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	50	1.735	13996.12	5.36	3344.74	2.36	3414.08	3.31	506.30	1.92	586.44	2.19	851.88	3.75
Absolute - Soft	100	1.723	14046.30	4.00	3375.64	2.68	2907.57	2.54	528.20	2.16	848.55	2.82	1118.26	2.19
FOSM	100	1.735	13929.07	3.30	3391.40	2.74	2378.00	1.83	526.00	2.12	1045.00	2.24	1508.00	1.87
SOSM	100	1.735	13930.10	3.29	3391.40	2.74	2378.40	1.78	526.00	2.12	1045.00	2.24	1507.60	1.88
Descending	100	1.737	13939.50	3.14	3375.10	2.72	2662.80	2.77	522.70	2.29	914.70	2.23	1313.70	2.56
1200	100	1.723	14044.21	4.15	3368.81	2.57	2959.63	3.78	525.62	2.08	827.53	3.86	1090.27	2.32
1300	100	1.723	14044.51	4.23	3367.82	2.44	2974.49	3.88	526.96	2.04	819.76	4.06	1082.57	2.8
1400	100	1.722	14050.05	3.99	3361.41	2.3	3017.02	2.49	526.83	2.09	794.97	3.12	1061.38	2.44
1500	100	1.723	14049.91	4.16	3348.86	2.52	3161.88	1.11	524.95	1.93	724.78	2.94	984.53	2.54
1600	100	1.729	14028.65	3.87	3337.43	2.79	3351.13	0.79	517.07	1.66	632.23	3.19	894.38	3.04

D.5. Results by Presence of Siblings

Table 14 Comparison Siblings vs. No-Siblings (MTB-F)

		Avg. Pref.		Assigned - Top [%]		Unassigned [%]	
		Mean	SE	Mean	SE	Mean	SE
Absolute - Hard	Siblings	1.977	0.002	0.368	0.001	0.208	0.001
	No-Siblings	1.886	0.0	0.453	0.0	0.174	0.0
Absolute - Soft	Siblings	1.978	0.001	0.356	0.001	0.221	0.001
	No-Siblings	1.871	0.0	0.458	0.0	0.17	0.0
FOSM	Siblings	2.029	0.001	0.343	0.001	0.228	0.001
	No-Siblings	1.867	0.0	0.461	0.0	0.168	0.0
SOSM	Siblings	2.029	0.001	0.343	0.001	0.228	0.001
	No-Siblings	1.867	0.0	0.461	0.0	0.168	0.0
Descending	Siblings	2.013	0.001	0.351	0.001	0.222	0.001
	No-Siblings	1.872	0.001	0.459	0.0	0.17	0.0
Ascending	Siblings	1.99	0.001	0.36	0.001	0.214	0.001
	No-Siblings	1.882	0.0	0.455	0.0	0.172	0.0

Appendix E: Extensions

E.1. Additional Groups

We discuss how the model presented in Section 3 can be easily extended to incorporate *secured-enrollment* and priorities obtained from siblings enrolled for the next academic year. Note that these are the only groups that have higher priority than those with siblings currently participating in the admissions process, so focusing on these is without loss of generality.

E.1.1. Secured Enrollment. Let $\bar{\mathcal{S}} \subseteq \mathcal{S}$ be the subset of students who were enrolled in some school in C during the most recent academic year, who can continue in the same school, and who are not seeking to switch schools in the current one. Note that a student $s \in \mathcal{S} \setminus \bar{\mathcal{S}}$ may also have been enrolled in some school during the most recent academic year; however, we assume that they are seeking to switch schools and would only prefer to be assigned to their current school if they cannot be assigned to a more preferred one.

To capture this, let $\bar{\mu}(s) \in C \cup \{\emptyset\}$ be the school where student $s \in \mathcal{S}$ was enrolled during the most recent academic year, with $\bar{\mu}(s) = \emptyset$ meaning that student s was not enrolled in any school during that time (e.g., s is applying to an entry-level or was enrolled in a private school that does not belong to C). Similarly, let $\bar{\mu}(c) \subseteq \mathcal{S}$ be the set of students who were enrolled in school $c \in C \cup \{\emptyset\}$ during the most recent academic year, and $\bar{\mu}^\ell(c) = \bar{\mu}(c) \cap \mathcal{S}^\ell$ for each $\ell \in \mathcal{L}$. Then, for each student $s \in \bar{\mathcal{S}}$, we assume that $\bar{\mu}(s) \in C$ and that they prefer $\bar{\mu}(s)$ over any other school in $C \cup \{\emptyset\} \setminus \{\bar{\mu}(s)\}$. In contrast, for each student $s \in \mathcal{S} \setminus \bar{\mathcal{S}}$, we assume that they strictly prefer at least one school in $C \cup \{\emptyset\} \setminus \{\bar{\mu}(s)\}$ over $\bar{\mu}(s)$. We formalize this in Assumption 1.[31]

ASSUMPTION 1. *For each student $s \in \bar{\mathcal{S}}$, $\bar{\mu}(s) \succ_s c$ for all $c \in C \cup \{\emptyset\} \setminus \{\bar{\mu}(s)\}$. In contrast, for each student $s \in \mathcal{S} \setminus \bar{\mathcal{S}}$, there exists at least one school $C \cup \{\emptyset\} \setminus \{\bar{\mu}(s)\}$ such that $c \succ_s \bar{\mu}(s)$.*

Another desirable property in any school choice system is that students who are currently enrolled have their seats secured and, consequently, cannot be displaced by other applicants unless assigned to a more preferred school. We refer to this property as *secured enrollment*.

DEFINITION 6 (SECURED ENROLLMENT). A matching μ satisfies *secured enrollment* if $\mu(s) \succeq_s \bar{\mu}(s)$ for all $s \in \mathcal{S}$.

In the remainder of this section, we assume that the clearinghouse aims to find a stable matching that satisfies *secured enrollment*. As a result, we assume that (i) each school c offers enough seats at each level to accommodate all students who have secured enrollment and that (ii) each student s with $\bar{\mu}(s) = c$ has higher priority than any student s' with $\bar{\mu}(s') \neq c$, i.e., $g(s,c) = 1$ if and only if $\bar{\mu}(s) = c$. We formalize this in Assumption 2.

ASSUMPTION 2. *For each school $c \in C$ and level $\ell \in \mathcal{L}$, $q_c^\ell \geq \left|\bar{\mu}^\ell(c)\right|$. Moreover, $g(s,c) = 1$ if and only if $\bar{\mu}(s) = c$ and, thus, $s \succ_c s'$ for all $(s,s') \in \bar{\mu}^\ell(c) \times \left(\mathcal{S}^\ell \setminus \bar{\mu}(c)\right)$.*

[31] If $\succ_s$ is such that $\emptyset \succ_s \bar{\mu}(s)$, then student s prefers to be unassigned over continue attending school $\bar{\mu}(s)$.

E.1.2. Priorities from Enrolled Siblings. As discussed in Section 1, most school districts consider sibling priorities by prioritizing students in schools where they have siblings enrolled during the most recent academic year and not seeking to switch schools. We formally define this type of priority, often called *static* sibling priority,[32] in Definition 7.

DEFINITION 7 (STATIC SIBLINGS PRIORITY). A student $s \in \mathcal{S}$ has *static* siblings priority in school $c \in C$ if $c \succ_s \bar{\mu}(s)$ and s has a sibling $s' \in f(s) \cap \bar{\mathcal{S}}$ such that $\bar{\mu}(s') = c$.

If there are no contingent priorities, incorporating secured enrollment and static priorities is straightforward, as the clearinghouse can preprocess the instance ensuring that (i) $g(s,c) = 1$ if and only if $\bar{\mu}(s) = c$, (ii) $g(s,c) = 2$ if and only if $\exists s' \in f(s) \setminus \{s\}$ such that $\bar{\mu}(s') = c$, and (iii) $g(s,c) = 3$ otherwise. Combined with a vector of random tie-breakers $\boldsymbol{p}$, these priority groups define a (unique) priority order $\succ_c$ for each school c, such that all students with secured enrollment have the highest priority, followed by students with static siblings priority, and finally, students with no priority. Then, the clearinghouse can proceed with the standard deferred acceptance algorithm Gale and Shapley (1962) to find a stable matching, as suggested below by Corollary 1. Note that Assumptions 1 and 2 ensure that all students with secured enrollment in c and who can continue in c and are not seeking to switch schools will get assigned to c for sure.

COROLLARY 1 **(Gale and Shapley (1962)).** *If school priorities are given by the combination of* static *priorities and random tie-breakers (with no contingent priorities), then a stable matching exists.*

The key distinction between *static* and *contingent* priorities is that the former depends on $\bar{\mu}$—which is invariant and known before the assignment—while the latter depends on the assignment μ. Given that students assigned to some school may decide not to enroll and, thus, the priority given may not be effective, it is natural to assume that students with static priorities prevail over students with contingent priorities. Indeed, this is the case in the Chilean school choice system, where siblings with static priority have the highest priority, and then students with contingent priority are considered only if there are vacancies left. We formalize this in Assumption 3.

ASSUMPTION 3. *Students with static priority have a higher priority than students with contingent priority.*

The formulations provided in Section 4 can be easily extended to account for static priorities under Assumption 3. In particular, the formulation for absolute priorities in (21) can be updated to account for more priority groups as follows:

$$\min \quad \sum_{(s,c) \in \mathcal{V}} r_{s,c} \cdot x_{s,c}$$

[32] When clear from the context, we will remove the word "sibling" from *static* (*contingent*, resp.) sibling priorities and simply refer to it as *static* (*contingent*, resp.) priorities.

$$s.t.\quad q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c}, \qquad \forall (s,c) \in \mathcal{V},\ g(s,c) < |\mathcal{G}| \tag{19a}$$

$$q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \succ_c s}} x_{a,c} + \sum_{\substack{f \in \mathcal{F}: \\ |f| \geq 2}} \sum_{\substack{\{a,a'\} \subseteq f: \\ a \in \mathcal{S}^{\ell(s)}, \\ a \prec_c s}} y_{a',a,c} + \sum_{\substack{a \in \mathcal{S}^{\ell(s)}: \\ a \prec_c s}} z_{a,c}$$
$$\forall (s,c) \in \mathcal{V},\ g(s,c) = |\mathcal{G}| \tag{19b}$$

$$x_{s',c} + \left(1 - \sum_{\substack{c' \in C: \\ c' \succeq_s c}} x_{s,c'}\right) \leq 2 - x_{a,c} + 1_{\{a \succ_c s\}} \cdot \left(z_{a,c} + \sum_{a' \in f(a) \setminus \{a\}} y_{a',a,c}\right),$$
$$\forall c \in C, f \in \mathcal{F}, \{s,s'\} \subseteq f, a \in \mathcal{S}^{\ell(s)} \setminus f,\ g(s,c) = g(a,c) = |\mathcal{G}| \tag{19c}$$

$$\mathbf{x} \in \mathcal{P},\ \mathbf{z} \in \mathcal{R}(\mathbf{x}),\ \mathbf{y} \in \mathcal{Q}(\mathbf{x},\mathbf{z}). \tag{19d}$$

Relative to (21), there two changes worth noticing. On the one hand, since students with contingent priority can only displace students with no priority, we update the constraints (19b) and (19c) to only consider students who initially have no priority, i.e., $g(s,c) = |\mathcal{G}|$. On the other hand, we know that students who initially belong to a higher priority group (i.e.., $g(s,c) < |\mathcal{G}|$), can only be displaced by students who have higher priority according to the initial order $\succ_c$, i.e., by students in a higher priority group ($g(s',c) < g(s,c)$) or students in the same group with a more favorable tie-breaker ($g(s',c) = g(s,c)$ and $p_{s',c} < p_{s,c}$). Thus, we add the set of constraints in (19a) to ensure that students with higher priority are not displaced by students with contingent priority. Note that this set of constraints is the analogous of the initial stability constraints in (3a).

REMARK 6. We can easily incorporate secured enrollment or any other group (e.g., students in the walk-zone, with parents working at the school, etc.) by redefining the initial groups. Furthermore, we can update the partial formulation and the soft counterparts analogously.

E.2. Maximize Siblings & Removing Random Tie-Breakers

As discussed in Section 5.3.2, two changes that leverage our optimization framework and that could increase the number of siblings assigned together are (i) changing the objective function from minimizing the sum of ranks of assignments to maximizing the number of receivers of priority, and (ii) removing the initial random tie-breakers to find the optimal assignment based solely on the priority groups. In what follows, we provide the details of each approach.

E.2.1. Changing the objective. To implement this, we replace the original objective function with the following:

$$\max \quad \sum_{\substack{f \in \mathcal{F} \\ |f| \geq 2}} \sum_{\{s,s'\} \subseteq f} \sum_{c \in C} y_{s,s',c}. \tag{20}$$

E.2.2. Removing initial tie-breakers. To implement this, in the *Rank* case, we can solve the following formulation:

$$
\begin{aligned}
\min \quad & \sum_{(s,c)\in\mathcal{V}} r_{s,c} \cdot x_{s,c} \\
s.t. \quad & q_c^{\ell(s)} \cdot \left(1 - \sum_{\substack{c'\in\mathcal{C}:\\ c'\succeq_s c}} x_{s,c'}\right) \leq \sum_{\substack{a\in\mathcal{S}^{\ell(s)}:\\ a\succeq_c s}} x_{a,c} + \sum_{\substack{f\in\mathcal{F}:\\ |f|\geq 2}} \sum_{\substack{\{a,a'\}\subseteq f:\\ a\in\mathcal{S}^{\ell(s)},\\ a\preceq_c s}} y_{a',a,c} + \sum_{\substack{a\in\mathcal{S}^{\ell(s)}:\\ a\preceq_c s}} z_{a,c}\,, \quad \forall (s,c)\in\mathcal{V}, && \text{(21a)}\\
& x_{s',c} + \left(1 - \sum_{\substack{c'\in\mathcal{C}:\\ c'\succeq_s c}} x_{s,c'}\right) \leq 2 - x_{a,c} + 1_{\{a\succeq_c s\}} \cdot \left(z_{a,c} + \sum_{a'\in f(a)\setminus\{a\}} y_{a',a,c}\right), \\
& \qquad\qquad\qquad\qquad \forall c\in\mathcal{C}, f\in\mathcal{F}, \{s,s'\}\subseteq f, a\in\mathcal{S}^{\ell(s)}\setminus f, && \text{(21b)}\\
& \mathbf{x}\in\mathcal{P},\ \mathbf{z}\in\mathcal{R}(\mathbf{x}),\ \mathbf{y}\in\mathcal{Q}(\mathbf{x},\mathbf{z}). && \text{(21c)}
\end{aligned}
$$

where $a \succ_c s$ ($a \succeq_c s$) implies that student a belongs to a priority group of higher (or equal, resp.) priority compared to that of student s at school c, i.e., $g(a,c) > g(s,c)$ ($g(a,c) \geq g(s,c)$, resp.). In the *Receivers* case, we simply modify the objective and use (20) instead.